When Does Value-Weighted Targeting Pay (and When It Does Not)?

A Diagnostic for Budget-Constrained Retention Targeting in Online Gaming

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Abstract

A marketer, a clinician, or a public-programme operator with a fixed budget can treat only a fraction of a heterogeneous population, where people differ both in how much they respond to treatment and in how much their response is worth. Practitioners routinely rank candidates by the product of two estimates: the treatment's effect on the outcome and the value of that outcome (for instance, lifetime value). We place this heuristic inside budget-constrained empirical welfare maximisation, where it is welfare-optimal as a corollary of the standard threshold result applied to a value-weighted outcome, and extend it with an operator-configurable preference weight. Throughout, "welfare" means operator welfare, distinct from the wellbeing of those targeted; the two can diverge.

The answer we reach is largely cautionary, and our contributions are diagnostic rather than a new estimator. Value weighting pays only at generous budgets: at a realistic campaign budget (reaching 10% of the active base) plain treatment-effect targeting matches or beats it, and a hybrid joint model (a joint treatment-effect model with a separate value model) outperforms the decomposed product rule at every budget we test. The operator-preference weight provably cannot raise welfare, so it prices restrictions rather than discovering them. What the decomposed framework adds is therefore not higher welfare but structure: a regret decomposition that identifies which estimated component is binding, and a map of when each component pays (value weighting helps under treatment-effect misspecification, but only when the value model carries the signal the treatment-effect model lost, and hurts at tight budgets). Because a player's value can be high precisely when they gamble in harmful volumes, we treat responsible-gaming safeguards as preconditions, not options. The main limitation is the absence of real-data validation, which caps how far the empirical map generalises.

Keywords: optimal treatment rules, causal inference, budget-constrained targeting, uplift modelling, policy learning, online gaming

1 Introduction

A decision-maker has a limited intervention (a retention bonus, a medical treatment, an educational programme) and a heterogeneous population. Each individual responds differently, is worth a different amount if they do respond, and has a different risk profile. The budget covers only a fraction of the population, and the decision-maker must choose which individuals to treat. This problem appears in personalised marketing (Lo, 2002; Hansotia and Rukstales, 2002), precision medicine, and public policy.

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Some notation. Write X for the vector of pre-treatment covariates observed for each individual (age, spend history, and so on), taking values in the covariate space $\mathcal{X}$. Write $Y \in \{0, 1\}$ for the binary outcome of interest (e.g., whether the individual is retained next period) and $W \in \{0, 1\}$ for the treatment indicator (1 if the individual received the intervention, 0 otherwise). Under the standard potential-outcomes framework, $Y(1)$ denotes the outcome the individual would realise if treated and $Y(0)$ the outcome they would realise if not treated; only one of these is ever observed for any given individual, which is why treatment effects must be estimated rather than measured directly. A subscript i indexes individuals, so X_i, Y_i, W_i refer to the covariates, outcome, and treatment assignment of individual i. A hat denotes an estimate: $\hat{\tau}(x)$ is an estimate of $\tau(x)$, and likewise for other quantities introduced below.

The standard approach estimates the conditional average treatment effect (CATE)

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x],$$

the expected gain in outcome from treating an individual with covariates x, and targets those with the highest estimates (Gutierrez and Gerardy, 2017; Künzel et al., 2019). Under a budget constraint the problem is one of optimal policy learning: choose a deterministic treatment rule $\pi : \mathcal{X} \to \{0, 1\}$ (a function that inputs an individual's covariates and outputs 1 if they should be treated and 0 otherwise) that maximises a *welfare function* $\mathcal{W}(\pi)$, subject to a cap $\kappa \in (0, 1)$ on the fraction of the population treated (Kitagawa and Tetenov, 2018; Athey and Wager, 2021). The welfare function is a scalar summary of how good a treatment rule is: loosely, the expected aggregate benefit produced when the rule is applied to the whole population, integrating over the distribution of covariates. In the canonical specification below, it is the expected treatment effect summed over the subset of the population the rule chooses to treat. The term is borrowed from econometrics and does not, on its own, refer to individual wellbeing. The canonical welfare function in this literature, empirical welfare maximisation (EWM), has two gaps that matter in practice. First, it treats every successful outcome as equally valuable: a low-value customer retained by treatment counts the same as a high-value regular. To fix this, this paper introduces an individual-level outcome value $V_i \geq 0$ (for example, customer lifetime value in a marketing setting, or quality-adjusted life-years in a medical one). Practitioners routinely multiply estimated uplift by customer lifetime value (LTV) to get a value-weighted ranking (Haupt and Lessmann, 2022; Gubela and Lessmann, 2021), but this heuristic has no welfare-theoretic justification. Second, EWM gives the decision-maker no formal mechanism for encoding domain-specific targeting preferences. Such preferences are common. They reflect operational priorities, regulatory requirements, or responsible-practice considerations, not claims about causal mechanisms. To encode them, the paper introduces an auxiliary risk covariate $R_i \in [0, 1]$ (for example, the individual's churn probability) and a preference weight $g(R_i)$ that the operator configures. In online gaming, for instance, an operator may prefer not to waste bonuses on stable players, and may want to limit intervention on very high-churn players who may be disengaging for welfare-relevant reasons (Ascarza, 2018). There is currently no principled way to encode such preferences inside the policy-learning objective, and no way to quantify what they cost.

A small example to fix ideas. Consider eight players with different treatment effects, lifetime values, and churn risk scores (see Example 1). Ranking by pure treatment effect hands a bonus to the player with the largest predicted uplift even when that player is almost certain to stay without it. The budget is wasted on a sure thing. Multiplying treatment effect by lifetime value fixes this, but can still send a bonus to a very stable player if their value is large enough. A third factor, configured by the operator, lets them impose an explicit veto: do not send bonuses to players below a churn threshold, no matter what the other numbers say. This paper places the product rule on a welfare-theoretic footing (it is the welfare-maximising rule under a budget constraint), but its more useful contributions are diagnostic and conditional: a

regret decomposition that tells the operator which of the three estimates is costing them welfare, a characterisation of when value weighting helps versus hurts, and a precise accounting of what the operator's preference factor costs when they choose to use it. One scope condition is load-bearing and worth stating up front: every optimality result here is conditional on the welfare function being *multiplicatively separable* in treatment effect, value, and preference weight (Definition 3). This is the form the $\hat{\tau} \cdot \hat{V}$ heuristic already takes, but it is an assumption, not a derivation, and Section 8 gives a plausible operator preference (willingness to tolerate risk that itself depends on value) under which it fails and the framework should not be used.

Contributions. We frame the paper deliberately as much around what does *not* pay as around what does: a unifying objective lets us state several cautionary results precisely (the operator-preference weight cannot improve welfare; a plain joint model beats the decomposition on welfare), and the positive contributions are a diagnostic and a map rather than a new optimal estimator.

1. **Welfare-theoretic placement of value-weighted targeting.** We show that ranking by estimated treatment effect times outcome value is the welfare-maximising rule under a budget constraint (Proposition 1), recovering the empirical-welfare-maximisation objective of Kitagawa and Tetenov (2018) and Athey and Wager (2021) when all values are equal. This connects the value-weighted decision rule studied in marketing (Haupt and Lessmann, 2022; Gubela and Lessmann, 2021; Hitsch and Misra, 2024) to budget-constrained policy learning. Neither value weighting (established by that prior work), nor budgeted ranking (established by the Qini/profit-uplift literature, Radcliffe, 2007; Devriendt et al., 2020), nor budget-constrained welfare-maximising assignment (established by Bhattacharya and Dupas, 2012) is itself new; our contribution is to combine outcome-value and operator-preference weighting in a single threshold/shadow-price characterisation within empirical welfare maximisation, together with the regret decomposition below.

2. **A regret decomposition that says which component to fix.** Welfare regret concentrates on individuals misranked near the targeting threshold (Proposition 2), and a telescoping identity attributes that regret to the estimation error in each component (Remark 9). This is the paper's main diagnostic: it tells an operator whether treatment-effect, value, or risk estimation is the binding constraint on welfare, and it explains the operating regimes below.

3. **When each component earns its place.** A simulation study identifies two regimes (Section 7): value weighting is most valuable when the treatment-effect model is misspecified *and* the value model happens to carry the dropped signal (an adversarial misspecification orthogonal to value removes the benefit and, at tight budgets, reverses it), and least valuable at very tight budgets (where it dilutes the treatment-effect signal). The recommendation to use the product rule is therefore budget- and model-quality-contingent, and we make the contingency explicit rather than leading with a single favourable budget.

4. **A tool for pricing targeting restrictions.** We add an operator-configurable preference weight on an auxiliary risk covariate and prove that any non-neutral setting weakly reduces operator welfare (Proposition 3). The practical output is a schedule of welfare costs for candidate restrictions, letting an operator price a governance or responsible-practice constraint before deploying it. We are explicit that welfare-based calibration of this weight returns the neutral setting by construction, so the mechanism prices preferences rather than discovering them.

We instantiate the framework for retention bonus targeting in online gaming, where the treatment effect is the causal impact of a bonus on retention, the outcome value is player lifetime

value, and the preference weight encodes the operator's view on where along the churn spectrum to concentrate interventions.

Roadmap. Section 2 reviews related work. Section 3 fixes notation, identification assumptions, and a notation reference table. Section 4 states and proves the composite-score result. Section 5 gives the regret decomposition and the welfare-cost bound for preference weighting. Section 6 instantiates the framework for online gaming with a piecewise linear preference function. Section 7 reports the simulation study. Sections 8 and 9 discuss scope and conclude.

2 Related Work

2.1 Policy Learning and Empirical Welfare Maximisation

The optimal treatment rule literature seeks to assign treatments so as to maximise a population-level welfare criterion. Manski (2004) establishes minimax regret bounds (bounds on the worst-case welfare loss relative to an oracle rule) for treatment choice under ambiguity. Kitagawa and Tetenov (2018) formulate empirical welfare maximisation (EWM) with the welfare function $\mathcal{W}(\pi) = \mathbb{E}[\pi(X_i) \cdot \tau(X_i)]$, which is simply the expected treatment effect summed over the subpopulation that the rule π chooses to treat, and a budget constraint $\mathbb{E}[\pi(X_i)] \leq \kappa$ that caps the fraction of the population treated at κ. Under these conditions, the optimal deterministic rule assigns treatment to individuals whose CATE exceeds a threshold determined by the budget. Athey and Wager (2021) develop doubly robust policy learning with $O(n^{-1/2})$ asymptotic regret guarantees under complexity constraints on the policy class (for example, finite VC dimension), meaning that welfare regret shrinks at a rate proportional to $1/\sqrt{n}$ as the sample size n grows. The rate is conditional on those complexity constraints; flexible policy classes such as unrestricted gradient-boosted tree ensembles do not directly satisfy them, and the practical rate in such settings may be slower. Their framework supports budget, fairness, and simplicity constraints. Budget-constrained welfare-maximising assignment itself predates the EWM formulation: Bhattacharya and Dupas (2012) characterise the welfare-maximising treatment assignment under an explicit budget (and more general linear) constraint, deriving the ranked-threshold structure directly from the constrained-optimisation problem. Our composite-score result is the outcome-value- and preference-weighted counterpart of theirs within the EWM/policy-learning framing, and the two-constraint extension of Corollary 2 sits alongside the broader constrained policy-learning literature that follows Athey and Wager (2021).

A parallel strand develops the estimation and evaluation machinery that any of these rules relies on. Kallus (2018) introduces balanced policy evaluation, providing efficient off-policy estimators that improve upon inverse propensity weighting for policy comparison. Zhao et al. (2012) develop outcome-weighted learning for estimating individualised treatment rules. Zhou et al. (2023) extend offline policy learning to the multi-action setting with generalisation guarantees. Chernozhukov et al. (2018) provide generic machine learning inference methods for heterogeneous treatment effects. Neither Kitagawa and Tetenov (2018) nor Athey and Wager (2021) includes an individual-level outcome-value weight in the welfare function or a mechanism for covariate-dependent targeting preferences; their objectives are recovered from ours by setting $V_i = 1$ for every individual i and $g \equiv 1$. Our welfare function nests theirs as a special case and extends it along both dimensions.

2.2 Risk-Averse Policy Learning

Manski and Tetenov (2007) study treatment rules for a risk-averse social planner whose preferences are a concave function of aggregate welfare (that is, the planner's utility falls faster than welfare does at the margin, so they dislike variance in outcomes). Our formulation differs. The

risk-modulation function $g(R_i)$, introduced in the preceding section, encodes preferences over an observable auxiliary covariate of the individual rather than over the distribution of aggregate welfare. This captures settings where the decision-maker has domain knowledge about which individuals are reachable by intervention, a common practical concern that existing risk-averse frameworks do not address.

2.3 Value-Weighted Uplift Targeting

Haupt and Lessmann (2022) study targeting under response-dependent costs and derive a profit-oriented decision rule in which candidates are ranked by a value- and cost-weighted treatment effect; ranking by $\hat{\tau}_i \cdot V_i$ (the product of estimated treatment effect and individual-level value) is the cost-homogeneous case. Here "uplift" is the marketing-literature term for the CATE, and V_i is the outcome value introduced above (typically customer lifetime value). Gubela and Lessmann (2021) propose value-driven evaluation metrics for uplift modelling that integrate treatment effect estimates with expected customer value, and the broader profit-uplift and profit-driven-analytics literature (Gubela et al., 2020; Fernández-Loría and Provost, 2022; Verbeke et al., 2017) frames targeting explicitly as expected-value (rather than effect) maximisation. Hitsch and Misra (2024) develop a decision-theoretic framework for heterogeneous targeting and characterise the optimal policy as ranking by expected incremental profit, i.e. effect weighted by margin or value. We do not claim to be the first to derive value-weighted targeting from a formal objective: this prior work already establishes its decision-theoretic merit. Our contribution is to place that rule inside the budget-constrained empirical-welfare-maximisation framework of Kitagawa and Tetenov (2018) and Athey and Wager (2021), where the optimal policy is a top-κ threshold rule (Proposition 1), and to extend it with the operator preference weight $g(R_i)$ and its welfare-cost accounting (Proposition 3).

The practice of allocating a fixed budget by ranking on an uplift-based score also has a long applied lineage in the Qini and profit-uplift literature (Radcliffe, 2007; Gubela et al., 2019; Devriendt et al., 2020), which evaluates budgeted targeting via cumulative-gain (Qini) curves and learns rankings directly for cost- or value-sensitive objectives. A budgeted top-κ cut of $\hat{\tau} \cdot \hat{V}$ is, operationally, a point on such a curve. We do not claim novelty for budgeted value ranking as a procedure; what the welfare-maximisation framing adds is (i) a precise optimality statement and threshold characterisation, (ii) the regret decomposition that attributes welfare loss to a specific component, and (iii) the preference weight g and its cost accounting. Devriendt et al. (2020) in particular formalise the observation, which our ordinal-sufficiency remark (Remark 8) also makes, that budgeted targeting depends only on the rank order of the score, not its calibrated magnitude.

2.4 Uplift Modelling

Uplift modelling provides the CATE estimates that enter our scoring rule. The framework is agnostic to the choice of CATE estimator. Any method that produces estimated individual treatment effects can serve as the uplift component, including T-Learners (which fit separate outcome models on the treated and control groups, then take their difference), S-Learners (which fit a single outcome model that takes treatment as an input feature), and X-Learners (which impute individual treatment effects before fitting a CATE model on those imputed values) (Künzel et al., 2019), causal forests (Wager and Athey, 2018), R-Learners (Nie and Wager, 2021), and doubly robust estimators (Kennedy, 2023).

2.5 Gaming Applications

Published applications of causal inference in online gaming remain limited relative to e-commerce and finance (Sifa et al., 2018; Braverman and Shaffer, 2012; Lemmens and Gupta, 2020); causal

(as opposed to predictive) targeting in gambling specifically is close to absent, which is part of the motivation for this paper. The broader retention-management literature (Ascarza et al., 2018; Lemmens and Gupta, 2020) establishes that targeting the highest-risk customers is often ineffective and that responsiveness, not risk, should drive allocation, which is precisely the structure our risk-modulation function and the moderation $m(p)$ encode. Responsible-gaming considerations increasingly require that targeting algorithms exclude vulnerable players from promotional offers, introducing a domain-specific constraint not present in standard uplift formulations. Devriendt et al. (2018) provide a comprehensive survey of uplift modelling methods applicable in such settings. Our risk-modulation function provides a principled mechanism for encoding such constraints within the welfare-optimisation framework.

3 Setup and Notation

Consider a population of N individuals indexed by $i \in \{1, \ldots, N\}$, each characterised by a feature vector $X_i \in \mathcal{X} \subseteq \mathbb{R}^d$, where the covariate space $\mathcal{X}$ is a subset of d-dimensional Euclidean space and d is the number of covariates measured per individual. Let $W_i \in \{0, 1\}$ be the binary treatment indicator (1 if treated, 0 otherwise) and $Y_i \in \{0, 1\}$ the binary outcome (e.g., retention).

Assumption 1 (Causal identification)**.** *The following standard conditions hold: (i)* unconfoundedness*: the potential outcomes are independent of treatment assignment given the covariates, $(Y_i(0), Y_i(1)) \perp W_i \mid X_i$ (once we condition on X_i, the treatment assignment carries no further information about counterfactual outcomes); (ii)* bounded overlap*: there exists a constant $\eta > 0$ such that $\eta \leq P(W_i = 1 \mid X_i = x) \leq 1 - \eta$ for every $x \in \mathcal{X}$, so that every individual has a non-trivial probability of receiving either treatment assignment; (iii)* SUTVA *(stable unit treatment value assumption): no interference between units, so that treating one individual does not affect another's outcome; (iv)* consistency*: $Y_i = Y_i(W_i)$, so that the observed outcome coincides with the potential outcome under the treatment actually received.*

Assumption 2 (Regularity conditions)**.** *(i) The composite score $s(X) = \tau(X) \cdot V(X) \cdot g(R(X))$ has a continuous distribution conditional on $g(R(X)) > 0$. This ensures that the threshold c^* in Proposition 1 is unique and that ties between individuals occur with probability zero. Note that the risk-modulation function g can create atoms (point masses) in the score distribution when $g(r) = 0$ over a set of positive probability; the continuity condition only applies to the non-zero portion of the score support. A bounded-constant plateau in g (for example, $g(r) = \phi > 0$ on $[p_H, 1]$ in the piecewise linear specification of Section 6) does not by itself create atoms in $s(X)$ as long as $\tau(X) \cdot V(X)$ is continuously distributed conditional on that plateau, because the product $\tau V \cdot \phi$ inherits the continuous distribution of τV. (ii) The expectation $\mathbb{E}[|\tau(X) \cdot V(X)|]$ is finite, so that the welfare function has a well-defined value (averaging treatment-effect times value does not blow up to infinity).*

Under Assumption 1, the conditional average treatment effect is identified:

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]. \tag{1}$$

In a randomised experiment where treatment is assigned independently of covariates, all four parts of Assumption 1 hold by design and $\tau(x)$ can be estimated from the observed data without further adjustment. In an observational setting (treatment assignment correlated with covariates) the same conditions still suffice in principle, but estimating $\tau(x)$ then requires modelling the propensity score $P(W_i = 1 \mid X_i)$ or using a treatment-effect estimator that absorbs confounding internally (see Section 8.3 for the corresponding welfare-evaluation machinery).

In addition to the CATE, the decision-maker observes two auxiliary quantities for each individual:

Definition 1 (Auxiliary components)**.** *Let $V(x)$ denote the* outcome value *of individual i (e.g., lifetime value, quality-adjusted life-years), and let $R(x) \in [0,1]$ denote the* risk covariate *(e.g., churn probability). Both are assumed to be pre-treatment quantities. We write $V(X_i) = V_i$ and $R(X_i) = R_i$ for notational convenience; $V(\cdot)$ and $R(\cdot)$ denote the underlying functions, while V_i and R_i denote their evaluation at individual i. We additionally assume $V(x) \geq 0$ for all x. This is a substantive restriction, not a normalisation: it rules out outcome-value metrics whose sign can flip for different individuals (for example, a harm-weighted welfare measure in a clinical trial where treatment is known to harm a subpopulation, so that V is negative for those individuals). The framework as stated applies to non-negative value metrics such as lifetime value or quality-adjusted life-years. Heterogeneous per-individual treatment* costs *are a separate concern, handled in Section 8 via the net-benefit score.*

Remark 1 (Conditional-mean reading of V and R)**.** *When V_i and R_i contain idiosyncratic noise beyond X_i (as in the simulated DGP of Section 7.1), $V(\cdot)$ and $R(\cdot)$ should be read as the conditional means $V(x) = \mathbb{E}[V_i \mid X_i = x]$ and $R(x) = \mathbb{E}[R_i \mid X_i = x]$. The welfare function* (3) *is invariant to this distinction by iterated expectations, and the oracle composite score throughout uses these conditional-mean quantities.*

Remark 2 (Pre-treatment requirement)**.** *If V_i or R_i reflects the effect of prior treatments (e.g., LTV elevated by previous promotional activity, churn risk modified by past interventions), conditioning on them introduces post-treatment bias and the causal interpretation of the composite score breaks down. This requirement constrains the auxiliary models to use only pre-treatment data.*

In operational practice the requirement is harder to satisfy than it looks. Customer lifetime value and churn probability in real systems are typically estimated from behavioural histories that were themselves shaped by prior treatments: spend patterns after previous bonuses, churn signals after previous campaigns, and so on. Strict compliance therefore requires either (i) segmenting individuals into cohorts defined before any promotional activity began, or (ii) partitioning the feature set of the auxiliary models to exclude any behavioural signal that post-dates treatment exposure. Practitioners deploying this framework on top of existing LTV and churn infrastructure should not assume their current models slot in cleanly. Their models may well include features that violate the pre-treatment requirement, and a careful audit of feature provenance is a necessary step before deployment.

A *deterministic treatment rule* is a function $\pi : \mathcal{X} \to \{0,1\}$ specifying which individuals receive treatment. The decision-maker faces a budget constraint:

Definition 2 (Budget-constrained targeting)**.** *Given budget proportion $\kappa \in (0,1)$, the feasible set of treatment rules is*

$$\Pi_\kappa = \{\pi : \mathcal{X} \to \{0,1\} \mid \mathbb{E}[\pi(X_i)] \leq \kappa\}. \tag{2}$$

Notation reference

Table 1 collects the symbols used throughout the paper. Section numbers indicate where each quantity is defined.

4 The Operator's Decision Problem

4.1 Welfare Function

We define the operator's objective as a generalised welfare function. Recall that a welfare function $\mathcal{W}(\pi)$ in this literature is a scalar summary of the expected aggregate benefit of applying the rule π to the population. The canonical welfare function of Kitagawa and Tetenov (2018)

Table 1: Notation reference.

Symbol	Meaning	First used
$X_i \in \mathcal{X}$	pre-treatment covariate vector for individual i	§3
$W_i \in \{0,1\}$	binary treatment indicator (subscripted; distinct from $\mathcal{W}(\pi)$)	§3
$Y_i \in \{0,1\}$	binary outcome (e.g., retention)	§3
$\tau(x)$	conditional average treatment effect (CATE)	§3
$V(x) \geq 0$	outcome value (e.g., lifetime value)	§3
$R(x) \in [0,1]$	auxiliary risk covariate (e.g., churn probability)	§3
$\pi : \mathcal{X} \to \{0,1\}$	deterministic treatment rule	§3
$\kappa \in (0,1)$	budget (fraction of population that may be treated)	§3
Π_κ	feasible set of treatment rules under budget κ	§3
$g : [0,1] \to [0,1]$	operator-configurable risk-modulation weight on R	§4
$\mathcal{W}(\pi; g)$	welfare of rule π: expected aggregate benefit, under preference g	§4
$s(x) = \tau(x)V(x)g(R(x))$	composite ranking score	§4
$h : \mathcal{X} \to [0,1]$	harm-protection weight on pre-treatment X	§6.2
$s^\dagger(x)$	harm-protected four-factor score, $\tau V g h$	§6.2
$s^\ddagger(x)$	harm-constrained penalised score, $\tau V g - \mu\,\text{harm}$	§6.2
c^*	welfare-optimal score threshold, satisfying the budget	§4
π^*	oracle-optimal rule; $\hat{\pi}$ its estimated counterpart	§4
$\mathcal{R}$	welfare regret of estimated rule (calligraphic; not risk covariate R)	§5
$\mathcal{M}$	misclassified set $\{i : \pi^*(X_i) \neq \hat{\pi}(X_i)\}$	§5
$\rho(\pi)$	risk exposure, mean R_i among treated	§5
$\theta = (p_L, p_{ML}, p_{MH}, p_H, \phi)$	parameters of the piecewise linear g	§6

sums treatment effects over the individuals the rule chooses to treat. The generalised version here weights each individual's treatment effect by two further quantities: the individual's outcome value (how valuable their response is) and an operator-configurable preference weight on an auxiliary risk covariate.

Definition 3 (Risk-modulated welfare)**.** *Let $g : [0,1] \to [0,1]$ be a* risk-modulation function*: a rule that takes a risk score $r \in [0,1]$ and returns a weight between 0 and 1. We require g to be measurable, a mild mathematical regularity condition ensuring that expectations involving g are well-defined; any sensible real-world function (step functions, piecewise linear functions, smooth functions) satisfies it. The non-negativity of g is substantive: it ensures that g does not flip the sign of the composite score (so a positive expected benefit cannot be turned into a negative one by the preference weight). The upper bound of 1 is a normalisation. Multiplying g by any positive constant leaves the ranking of individuals unchanged, so we lose no generality by capping g at 1. The risk-modulated welfare of a treatment rule π is*

$$\mathcal{W}(\pi; g) = \mathbb{E}\big[\pi(X_i) \cdot \tau(X_i) \cdot V_i \cdot g(R_i)\big]. \tag{3}$$

The multiplicative form is a modelling assumption: it requires that the operator's valuation of treating individual i is separable into the three components. We discuss the scope and limitations of this assumption in Section 8. When $g \neq 1$, the operator is expressing a preference for concentrating treatment on specific risk segments, accepting a reduction in $\mathcal{W}(\pi; \mathbf{1})$ (the unmodulated welfare) in exchange for targeting properties they value. The function g thus plays a different role from τ and V: the latter two are estimable quantities about the world, while g encodes the operator's preferences about where to intervene.

The welfare function (3) has a direct interpretation: it is the expected total value generated by treatment, where each individual's contribution is their treatment effect (τ), weighted by how valuable their response is (V), and further modulated by the operator's preference weight for their risk profile (g).

Remark 3 (Nesting existing frameworks)**.** *Setting $V_i = 1$ and $g \equiv 1$ recovers the standard EWM welfare of Kitagawa and Tetenov (2018): $\mathcal{W}(\pi) = \mathbb{E}[\pi(X_i) \cdot \tau(X_i)]$. Setting $g \equiv 1$ alone*

recovers value-weighted welfare: $\mathcal{W}(\pi) = \mathbb{E}[\pi(X_i) \cdot \tau(X_i) \cdot V_i]$, the implicit objective underlying the empirical practice of Haupt and Lessmann (2022) and Gubela and Lessmann (2021). The full formulation introduces the risk-modulation function g as the novel component.

4.2 Optimal Treatment Rule

Proposition 1 (Optimal risk-modulated treatment rule)**.** *Under Assumptions 1 and 2, the treatment rule that maximises $\mathcal{W}(\pi; g)$ subject to $\pi \in \Pi_\kappa$ is*

$$\pi^*(x) = \mathbf{1}\{\tau(x) \cdot V(x) \cdot g(R(x)) > c^*\}, \tag{4}$$

where c^* is the unique threshold satisfying $\mathbb{E}[\pi^*(X_i)] = \kappa$. Equivalently, π^* targets the top-κ fraction of individuals ranked by the composite score

$$s(x) = \tau(x) \cdot V(x) \cdot g(R(x)). \tag{5}$$

Proof. A Lagrangian on the budget is maximised pointwise: treat exactly when $\tau(x)V(x)g(R(x)) > \lambda$. Strong duality for the convex relaxation (Slater's condition, with the measure-zero boundary of Assumption 2(i)) makes the relaxed optimum binary almost surely, and $c^* = \lambda^*$ exhausts the budget. This is the Neyman-Pearson threshold logic of Kitagawa and Tetenov (2018, Theorem 1) applied to the three-factor product; the full argument, including the slack-budget case, is in Appendix A. □

Remark 4 (Reduction to a transformed outcome)**.** *Proposition 1 is the standard empirical-welfare-maximisation threshold result of Kitagawa and Tetenov (2018), made exact by a change of outcome variable rather than a new argument. Define the transformed outcome $\tilde{Y}_i = Y_i \cdot V_i \cdot g(R_i)$. Because V_i and $g(R_i)$ are pre-treatment functions of X_i (Definition 1), they pass through the conditional-expectation operator, so the CATE of the transformed outcome is*

$$\tilde{\tau}(x) = \mathbb{E}[\tilde{Y}(1) - \tilde{Y}(0) \mid X = x] = \tau(x) \cdot V(x) \cdot g(R(x)) = s(x).$$

The risk-modulated welfare (3) is then exactly the unweighted EWM welfare of Kitagawa and Tetenov (2018) applied to $\tilde{Y}$, and Proposition 1 is their Theorem 1 for that outcome verbatim. One caveat on the per-x identity: when V_i and R_i carry idiosyncratic noise beyond X_i (the conditional-mean reading of Remark 1), $\tilde{\tau}(x) = \mathbb{E}[(Y(1) - Y(0))\, V\, g(R) \mid X = x]$ equals $\tau(x)\, V(x)\, g(R(x))$ exactly only when the treatment effect is conditionally independent of the weights and $\mathrm{Cov}(V, g(R) \mid X) = 0$; otherwise it carries an additional conditional-covariance term. The welfare function itself is invariant to this (by iterated expectations, as in Remark 1), so the optimality of the ranking is unaffected; the product form is the exact transformed-CATE only under the conditional-independence reading used throughout. We state the result in product form, and treat the reduction as the point rather than the optimality, because the decomposition into (τ, V, g) is what the rest of the paper exploits: the three factors are estimated, validated, and updated component by component (Remark 7), and the regret decomposition (Proposition 2) attributes welfare loss to individual factors. The same reduction explains the end-to-end joint model of Section 7.7: a single model trained to predict the value-weighted outcome estimates $\tilde{\tau} = s$ directly, which is precisely the object the welfare function ranks on, and is why it is competitive with the decomposition without ever forming the product explicitly.

Remark 5 (Interpretation of the threshold)**.** *The threshold c^* has a simple economic meaning: it is the minimum composite score at which it is worth spending a unit of budget on an individual. Anyone with a score above c^* is treated; anyone below is not. In economic terminology, c^* is the* shadow price *of the budget: the welfare that a marginal unit of additional budget would buy. As the budget grows ($\kappa \to 1$), more individuals clear the bar, so c^* falls toward the lowest composite score present in the population (the* infimum *of the score distribution). In the unconstrained limit, every individual with a positive composite score is treated.*

Remark 6 (What g encodes)**.** *The function g is a mechanism for the operator to express a preference about where along the auxiliary-covariate distribution they want to intervene. It enters the welfare function multiplicatively, which has three consequences. First, setting $g(r) = 0$ for some values of r places zero weight on treating individuals with that risk profile no matter what their treatment effect or value; this veto property lets the operator encode hard exclusions (e.g., do not target very low-churn players) directly in the objective rather than as an external post-hoc filter. Second, because the welfare function is multiplicative, g is distinct in character from τ and V: the latter two are estimable quantities about the world; g is a preference, not a claim about causal mechanisms. Third, an operator who chooses $g \neq 1$ accepts a quantifiable welfare cost (Proposition 3) in exchange for the targeting pattern they prefer. Even a perfectly specified CATE model does not foreclose choosing $g \neq 1$; that choice just has to be justified on grounds other than welfare.*

4.3 Decomposability and Ordinal Sufficiency

Remark 7 (Decomposability)**.** *The proof is a standard Lagrangian argument, so the result's contribution is the decomposition rather than the optimality per se. The composite score $s(x) = \tau(x) \cdot V(x) \cdot g(R(x))$ separates the targeting decision into three components that can be estimated, validated, and updated independently. If the operator replaces any single component while holding the others fixed, the resulting score remains welfare-maximising for the correspondingly modified welfare function; this follows from the multiplicative structure of* (3)*. The operator can audit the uplift model, the value estimates, and the risk-modulation function separately, diagnose failures in any one of them, and swap in an improved version without retraining the others. A single-model policy learner does not have this property and must be retrained end-to-end when any aspect of the targeting objective changes.*

Remark 8 (Ordinal sufficiency)**.** *The top-κ targeting rule only cares about the rank ordering of individuals by composite score, not about the scores' exact magnitudes. (Ranking "by order" rather than by "size" is the sense in which the rule is* ordinal *rather than* cardinal*.) So if an estimated score $\hat{s}(x) = \hat{\tau}(x) \cdot \hat{V}(x) \cdot \hat{g}(\hat{R}(x))$ happens to rank individuals in the same order as the true (oracle) score $s(x)$, then the estimated rule picks exactly the same set of people as the oracle rule. Formally, $s(x) > s(x') \Leftrightarrow \hat{s}(x) > \hat{s}(x')$ for all pairs implies the top-κ sets coincide. Rank preservation is, however, a strong condition, especially for a product of three estimated components. Even if each component's estimator is monotone in the true component (for example, $\hat{\tau}(x) = h_\tau(\tau(x))$ for some strictly increasing h_τ), the product $h_\tau(\tau) \cdot h_V(V)$ may rank individuals differently from the true product $\tau \cdot V$ when h_τ and h_V are different transformations. A simple example: suppose $\hat{\tau} = \tau^2$ and $\hat{V} = V$. An individual with $(\tau = 1, V = 10)$ and an individual with $(\tau = 2, V = 3)$ have true products $1 \cdot 10 = 10$ and $2 \cdot 3 = 6$, so the first ranks higher. Their estimated products are $1^2 \cdot 10 = 10$ and $2^2 \cdot 3 = 12$, reversing the order. When ranks are not perfectly preserved, the welfare loss depends on how much oracle-score mass sits on the set of misranked individuals near the targeting threshold (Proposition 2 makes this precise).*

5 Properties

5.1 Welfare Regret Decomposition

We now ask: if the operator uses estimates $(\hat{\tau}, \hat{V}, \hat{g})$ instead of the true (but unknown) quantities (τ, V, g), how much welfare does the resulting rule leave on the table? The gap between the oracle rule (which uses the truth) and the estimated rule is the *welfare regret*. Call the oracle rule π^* and the estimated rule $\hat{\pi}$, and call the regret $\mathcal{R} = \mathcal{W}(\pi^*; g) - \mathcal{W}(\hat{\pi}; g)$.

The result below is a *structural decomposition*: it rewrites the regret in a form that reveals where the loss comes from, rather than a *statistical convergence bound* that tells us how fast the regret

shrinks as we collect more data. Both are useful; they answer different questions.

Proposition 2 (Welfare regret decomposition)**.** *Let π^* denote the oracle-optimal rule (which uses true τ, V, g) and $\hat{\pi}$ the estimated rule (which uses $\hat{\tau}$, $\hat{V}$, $\hat{g}$). Define the regret $\mathcal{R} = \mathcal{W}(\pi^*; g) - \mathcal{W}(\hat{\pi}; g)$. Then*

$$\mathcal{R} = \mathbb{E}\big[(\pi^*(X_i) - \hat{\pi}(X_i)) \cdot \tau(X_i) \cdot V_i \cdot g(R_i)\big] = \mathbb{E}\big[(\pi^*(X_i) - \hat{\pi}(X_i)) \cdot s(X_i)\big], \tag{6}$$

where $s(x) = \tau(x) \cdot V(x) \cdot g(R(x))$ is the oracle composite score. Intuitively: the regret only picks up contributions from individuals the two rules disagree on. Call this the misclassified set*: $\mathcal{M} = \{i : \pi^*(X_i) \neq \hat{\pi}(X_i)\}$. The regret is bounded by the average oracle score on that set:*

$$|\mathcal{R}| \leq \mathbb{E}\big[|s(X_i)| \cdot \mathbf{1}\{i \in \mathcal{M}\}\big]. \tag{7}$$

When both rules exactly exhaust the budget ($\mathbb{E}[\pi^*] = \mathbb{E}[\hat{\pi}] = \kappa$), each false negative ($\pi^* = 1$, $\hat{\pi} = 0$) is paired with a false positive ($\pi^* = 0$, $\hat{\pi} = 1$), and the c^* contributions cancel, giving the tighter bound

$$\mathcal{R} = \mathbb{E}\big[|s(X_i) - c^*| \cdot \mathbf{1}\{i \in \mathcal{M}\}\big], \tag{8}$$

which shows that misclassification hurts welfare in proportion to how far each misclassified score sits from the threshold, not in proportion to its absolute magnitude. In words: estimation mistakes only hurt welfare to the extent that the mistaken individuals are near the margin of being worth treating. The budget-exhaustion condition holds automatically for top-κ-percentile selection on a continuously distributed score, which is the rule used by π^* and by every estimated strategy in Section 7; if instead $\hat{\pi}$ is obtained by thresholding an estimated score and the threshold does not exhaust the budget, the looser bound (7) *still applies.*

Proof. By linearity of expectation the regret reduces to $\mathbb{E}[(\pi^*(X_i) - \hat{\pi}(X_i)) \cdot s(X_i)]$, an expectation over the misclassified set $\mathcal{M}$ alone (the rules agree elsewhere); bounding $|\pi^* - \hat{\pi}| \leq 1$ gives (7). When both rules exhaust the budget the false-negative and false-positive masses are equal, so the c^* terms cancel and the tighter $|s - c^*|$ form (8) follows. Full derivation in Appendix A. □

Remark 9 (Component-wise score error)**.** *The bound* (7) *tells us that regret depends on the oracle-score magnitude on the misclassified set. A related question is* why *individuals end up misclassified, and what to do about it. The score estimation error for a single individual is $|s(x) - \hat{s}(x)| = |\tau(x)V(x)g(R(x)) - \hat{\tau}(x)\hat{V}(x)\hat{g}(\hat{R}(x))|$. Applying the* triangle inequality *(the elementary fact that $|a + b| \leq |a| + |b|$) after replacing oracle components with their estimates one at a time gives the bound*

$$|abc - \hat{a}\hat{b}\hat{c}| \leq |a - \hat{a}| \cdot |b| \cdot |c| \;+\; |\hat{a}| \cdot |b - \hat{b}| \cdot |c| \;+\; |\hat{a}| \cdot |\hat{b}| \cdot |c - \hat{c}|.$$

This manoeuvre is called a telescoping argument *(each replacement extends the "telescope" one section). The bound decomposes the total score error into three terms, one per component. It is diagnostic: it tells the operator which component's estimation error contributes most to misranking at the targeting threshold. Errors in components with large magnitude (for example, outcome value V, which can vary by orders of magnitude when V is customer lifetime value) are amplified by the magnitudes of the other two components. The risk-modulation function g, being bounded in $[0, 1]$, has limited scope to amplify errors in the other components.*

Remark 10 (Comparison with statistical regret bounds)**.** *Our regret decomposition answers a different question from the finite-sample regret bounds of Athey and Wager (2021). Athey and Wager establish that under their empirical welfare maximisation setup, welfare regret shrinks at rate $O(n^{-1/2})$ as sample size n grows (so doubling the sample roughly halves the square-rooted regret). That is a* statistical *convergence rate: how welfare regret shrinks with more data. Proposition 2 is* structural*: given any finite sample, it decomposes the regret into (a) the size of*

the set where the estimated rule disagrees with the oracle and (b) the oracle score mass on that set, identifying which component's estimation error is doing the damage. The two results are complementary, not competing.

5.2 The Value-Conservatism Trade-off

The operator's preference weight g is a dial they can turn. Turning it away from the neutral setting ($g \equiv 1$, i.e., weight every individual equally regardless of risk) lets them exclude segments or concentrate treatment on others. The question is: what does turning that dial cost?

We measure cost in *unmodulated welfare*, written $\mathcal{W}(\pi; \mathbf{1})$: welfare evaluated as if the preference weight were neutral even when the operator chose a non-neutral one. This is the measure that tells us how much benefit the operator's chosen rule actually generates for the population, ignoring how the operator feels about the distribution of that benefit. Proposition 3 answers it with a bound that follows by construction: any non-neutral preference weakly reduces unmodulated welfare, because π_0^* maximises unmodulated welfare by definition and any other feasible rule is weakly worse. The proposition is therefore best read as a consistency check, not a discovery. Its practical value is indirect: paired with the empirical cost schedule of Section 7.6, it lets the operator make an informed decision about whether the targeting control they want is worth its welfare cost.

Definition 4 (Risk exposure)**.** *The risk exposure of a treatment rule π is the expected value of the risk covariate among treated individuals:*

$$\rho(\pi) = \mathbb{E}\big[R_i \mid \pi(X_i) = 1\big]. \tag{9}$$

Proposition 3 (Value-conservatism trade-off)**.** *Under Assumptions 1 and 2, let π_0^* denote the optimal rule under $g_0 \equiv 1$ (no risk modulation) and π_α^* the optimal rule under a risk-modulation function g_α with $0 \le g_\alpha(r) \le 1$ for all r. Then the unrestricted welfare satisfies*

$$\mathcal{W}(\pi_\alpha^*; \mathbf{1}) \le \mathcal{W}(\pi_0^*; \mathbf{1}). \tag{10}$$

That is, introducing any non-trivial risk modulation ($g_\alpha \not\equiv 1$) weakly reduces the unrestricted welfare relative to the unmodulated optimum. Equality holds if and only if g_α does not change the top-κ set: the set of individuals targeted by π_α^* is identical to the set targeted by π_0^*.

Proof. The rule π_0^* maximises $\mathcal{W}(\pi; \mathbf{1}) = \mathbb{E}[\pi(X_i) \cdot \tau(X_i) \cdot V_i]$ over Π_κ by construction. The rule π_α^* maximises a different objective, $\mathcal{W}(\pi; g_\alpha) = \mathbb{E}[\pi(X_i) \cdot \tau(X_i) \cdot V_i \cdot g_\alpha(R_i)]$. Since $\pi_\alpha^* \in \Pi_\kappa$ is feasible for the unrestricted problem but is not in general optimal for it, we have $\mathcal{W}(\pi_\alpha^*; \mathbf{1}) \le \mathcal{W}(\pi_0^*; \mathbf{1})$. For the equality condition: $\mathcal{W}(\pi_\alpha^*; \mathbf{1}) = \mathcal{W}(\pi_0^*; \mathbf{1})$ if and only if π_α^* is also optimal for the unrestricted welfare, which under Assumption 2 (unique threshold) holds if and only if the top-κ sets coincide. □

Remark 11 (Welfare-risk frontier)**.** *As the operator varies the preference function g, each choice produces a different pair (welfare, risk exposure). Plotting all such pairs traces out a curve in the welfare-risk plane, analogous to a Pareto frontier in multi-objective optimisation (a curve showing the best achievable value on one objective for each level of the other). Proposition 3 guarantees that the unmodulated rule ($g \equiv 1$) sits at the highest-welfare point on this curve. It does not guarantee that the curve is monotone in any particular dial the operator turns; a small change to g's shape can move both welfare and risk exposure in complicated ways.*

Remark 12 (When g can offset estimation error)**.** *Proposition 3 is a statement about oracle-optimal rules (rules that use the truth). Once we switch to using estimates, the story is not quite as clean. Suppose the true treatment effect factors multiplicatively as $\tau(x) = \tau_0(x) \cdot m(R(x))$,*

where τ_0 is a "base" effect that depends on covariates other than the risk variable, and $m : [0,1] \to [0,1]$ is a moderation function *that says how the treatment effect is attenuated by the risk covariate (for instance, a treatment is most effective at moderate risk and vanishes at the extremes). Suppose further that the estimated treatment-effect model cannot capture m (because its feature set or flexibility is too limited). Then inserting $g = m$ into the composite score yields $\hat{\tau}(x) \cdot \hat{V}(x) \cdot m(\hat{R}(x))$, which approximates the true ranking to the extent that each estimated component tracks the rank contribution of its oracle counterpart. In this regime, the estimated rule that uses a non-neutral g may beat the estimated rule that uses a neutral g, because the operator's preference factor is silently compensating for what the treatment-effect model missed. This is a secondary use of the framework, not its primary justification: it only applies when g happens to mirror a real structural interaction the treatment-effect model missed, and the compensation is partial because m itself has to be approximated. The simulation in Section 7 does not exhibit this effect: even under the CATE misspecification of Section 7.3, calibration selects $g \approx 1$. The reason is that the misspecification tested there is feature dropout in the CATE model, which the value model partially absorbs through features it retains; this is outside the regime described above, which requires the operator to supply a g that mirrors a structural interaction no estimated component can recover. The Remark's regime is therefore a theoretical case in which g can help, not a claim about the DGP used in this paper, and the welfare improvement observed in Section 7.3 comes from value weighting, not from g.*

6 Instantiation: Online Gaming

We now instantiate the general framework for retention bonus targeting in online gaming. By "online gaming" we mean the higher-frequency, variable-stake end of the sector (iGaming-style casino and slots play), where players engage repeatedly, spend varies over orders of magnitude, and retention and churn are the natural outcome and risk covariate; the simulated data-generating process of Section 7.1 reflects that vertical, with its log-normal weekly spend and multi-session login cadence. Lower-frequency, draw-anchored products such as lottery differ in two ways that matter for the framework: play cadence is slower (so a "retention window" spans draws rather than sessions) and the value distribution is less extreme-tailed. The heavy-tailed-value robustness check in Section 7.7 brackets the first of these; an operator in a lower-tail vertical should expect the value-weighting effects reported here to be milder, since the paper's own analysis shows heavier value tails sharpen them.

The treatment is a binary retention bonus ($W_i = 1$: bonus received; $W_i = 0$: no bonus). The outcome is binary retention ($Y_i = 1$: player remains active). The retention window (the period over which "active" is measured, e.g., 7 days, 30 days, or 90 days) is an operator choice; all results hold for whichever window is used, but operators should note that treatment effects can differ across windows (a bonus that retains for a week may accelerate longer-term churn through habituation), and cross-window effects are a separate modelling concern beyond the scope of this framework. The outcome value is player lifetime value ($V_i = \text{LTV}_i$). The risk covariate is churn probability ($R_i = p_i^{\text{churn}}$). Throughout this section and Section 7 we use p in place of R for the risk covariate, to keep the notation close to the domain convention of referring to churn probability as p; the symbols denote the same quantity.

6.1 The Risk-Modulation Function

The effectiveness of a retention bonus depends on the player's position along the churn risk spectrum. Domain reasoning motivates three operational regimes:

1. **Low churn risk** ($p < p_L$). The player is stable and would likely remain active without intervention. A bonus consumes budget without generating incremental retention. The

risk-modulation weight should be zero.

2. **Moderate churn risk** ($p \in [p_L, p_H]$). The player is at risk but reachable; a well-timed bonus can plausibly tip the retention decision. The weight should be at its maximum.

3. **High churn risk** ($p > p_H$). The player has likely disengaged to the point where a bonus cannot reverse the trajectory (Ascarza, 2018). The weight should be reduced, though a non-zero floor acknowledges uncertainty in the churn estimate.

We use a five-parameter piecewise linear g that encodes these regimes with smooth transitions. Because the specific functional form is an implementation choice rather than part of the contribution, we defer the formal definition and its regularity (Lipschitz) properties to Appendix B and keep only the operational reading here. The parameters $\theta = (p_L, p_{ML}, p_{MH}, p_H, \phi)$ have direct operational meanings: p_L is the churn probability below which players are excluded (budget conservation); p_{ML} and p_{MH} bound the "fully reachable" plateau where the weight is one; p_H is the churn probability above which the weight stops falling; and ϕ is the floor weight for the highest-risk players. Figure 1 plots g at the default values $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$ used in the simulation. These defaults are pedagogical (chosen to produce visually clear contrasts across the regimes), not operational recommendations; real operators should set θ from domain evidence (for instance, p_L tends to be higher, 0.10 to 0.20, in real campaigns because "very stable player" is a commercial judgment, not a statistical one). The frontier analysis in Section 7.6 reports welfare costs for four alternative configurations (wide, default, narrow, aggressive); an operator can reproduce the same analysis for whichever θ is under consideration and read off the corresponding welfare cost.

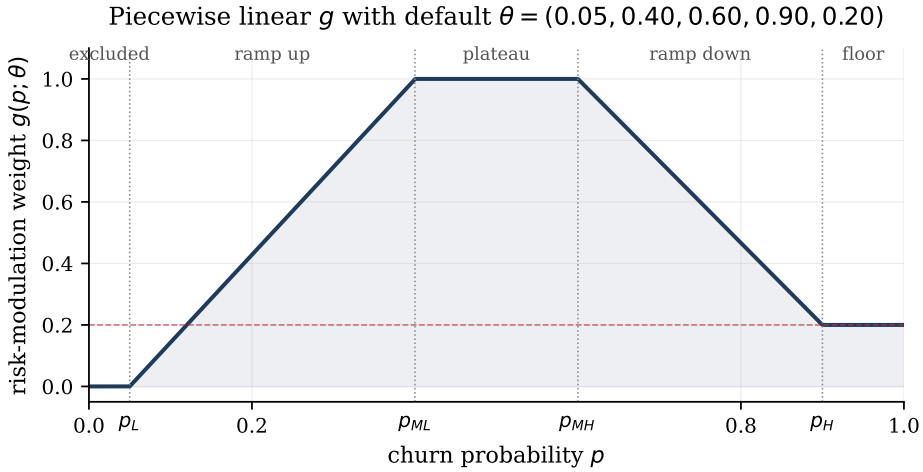


Figure 1: The piecewise linear risk-modulation function at default parameters $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$. Players below p_L are excluded. Weight ramps to one through the moderate-risk plateau and falls to the floor ϕ for the highest-churn players. Formal definition in Appendix B.

6.2 Responsible-Gaming Considerations

The framework supports a pre-targeting *exclusion set* $\mathcal{E}$: a pre-specified list of individuals who are removed from the pool before ranking, regardless of their composite score. Writing $\mathcal{S}^*$ for the final set of individuals treated, the rule becomes "apply top-k ranking among individuals not in $\mathcal{E}$", or in notation $\mathcal{S}^* = \arg\text{top-k}_{i \notin \mathcal{E}}\, s_i$. In the gaming context, $\mathcal{E}$ should be determined

by validated vulnerability screening instruments aligned with jurisdiction-specific regulatory requirements (e.g., the Problem Gambling Severity Index and self-exclusion registries in most jurisdictions; affordability checks and Gambling Commission licensing conditions in the United Kingdom; operator-side constraints from advertising regulators such as the Advertising Standards Authority or comparable bodies in other markets). This exclusion operates *above* the scoring framework: the welfare-optimal rule from Proposition 1 applies to the eligible population $\{1, \ldots, N\} \setminus \mathcal{E}$.

The floor parameter $\phi > 0$ means the framework assigns non-zero targeting weight to the highest-churn players. Some of these players may be churning because gambling is causing them harm; a production system should incorporate a "should we intervene?" gate informed by harm indicators beyond the scope of this framework, prior to the "who should we target?" ranking. The risk-modulation function encodes the operator's *targeting preferences*, not claims about *player welfare*. Setting $p_L > 0$ excludes stable players (budget conservation); reducing ϕ toward zero limits intervention on very high-churn players, some of whom may be disengaging for welfare-relevant reasons. The overlap is a side-effect of an operator-preference setting, not a principled harm gate: responsibility for harm-sensitive exclusions sits with $\mathcal{E}$ and the harm-indicator safeguards discussed below. However, the framework does not replace dedicated responsible-gaming safeguards: the exclusion set $\mathcal{E}$ requires domain expertise and regulatory alignment beyond the scope of this paper.

A related concern deserves its own flag. The V-weighting step, which is central to the paper's welfare-optimal result, multiplies the treatment-effect score by customer lifetime value. In online gaming this can compound harm: a player's lifetime value can be large precisely because they are gambling in harmful volumes, and value-weighted targeting will then concentrate bonuses on exactly those players. Nothing in the composite score detects this. We therefore treat two safeguards as preconditions, not options, for any responsible-gaming deployment of this framework: (a) the exclusion set $\mathcal{E}$, populated from validated vulnerability signals, and (b) a harm gate that operates prior to (or in addition to) $\mathcal{E}$, using behavioural harm indicators (deposit velocity, session duration escalation, loss-chasing patterns) that are outside the scope of this framework. The composite-score ranking should not be deployed without both. Two further deployment defaults follow. First, the system should *fail closed*: if harm signals are unavailable for part of the population, the high-value tail of that part should be excluded by default rather than scored unguarded, so that missing harm data never silently reverts the rule to the unprotected score $\tau V g$. Second, membership of $\mathcal{E}$ and the harm signals feeding h must be re-evaluated at each campaign, not cached: vulnerability is dynamic, and a player outside $\mathcal{E}$ at one scoring round may have crossed a harm threshold by the next. Relatedly, although we use a non-zero floor $\phi = 0.20$ in the simulation for illustrative contrast, we recommend a default of $\phi = 0$ (no targeting weight on the highest-churn players) for live deployment, with any positive ϕ requiring explicit justification, since the highest-churn players are the most likely to be disengaging for welfare-relevant reasons; in some jurisdictions, direct promotion to identified at-risk players may also be restricted, which independently favours $\phi = 0$. Two ways to operationalise this within the existing machinery: (i) cap V_i above a threshold calibrated against harm indicators, so that the composite score stops growing with value once harm signals are present; or (ii) multiply V_i by a separate harm-protection factor that downweights individuals whose high value correlates with harm signals, logically distinct from $g(R_i)$ because it operates on the value dimension rather than the risk dimension. Both approaches admit a clean formal treatment as a fourth multiplicative factor in the welfare function, so harm protection can be an integral component of the ranking rather than an adjacent override.

Corollary 1 (Harm-protected welfare-optimal rule)**.** *Let $h : \mathcal{X} \to [0, 1]$ be a measurable* harm-protection weight *defined on the pre-treatment covariate vector, and define the extended welfare*

function

$$\mathcal{W}(\pi; g, h) = \mathbb{E}\big[\pi(X_i) \cdot \tau(X_i) \cdot V_i \cdot g(R_i) \cdot h(X_i)\big]. \tag{11}$$

Under Assumptions 1 and 2 (with the regularity condition applied to the four-factor score), the treatment rule that maximises $\mathcal{W}(\pi; g, h)$ subject to $\pi \in \Pi_\kappa$ ranks individuals by the four-factor composite score $s^\dagger(x) = \tau(x) \cdot V(x) \cdot g(R(x)) \cdot h(x)$. Further, if $\pi^*_{g,h}$ denotes the welfare-optimal rule for given (g, h) and $\pi^*_{1,1}$ the rule under $g \equiv 1, h \equiv 1$, then

$$\mathcal{W}(\pi^*_{g,h}; \mathbf{1}, \mathbf{1}) \;\le\; \mathcal{W}(\pi^*_{1,1}; \mathbf{1}, \mathbf{1}), \tag{12}$$

with equality if and only if the top-κ sets coincide.

Proof. Both claims follow from the proofs of Propositions 1 and 3 applied to the four-factor integrand: the Lagrangian argument pins π^* down pointwise on $s^\dagger(x)$, and the feasibility argument for (12) mirrors that of Proposition 3 with $g \cdot h$ in place of g_α. □

Value capping is the special case of Corollary 1 in which $h(x)$ equals $\bar{V}/V(x)$ on the capped set and 1 elsewhere, so the product $V(x) \cdot h(x)$ equals $\min(V(x), \bar{V})$; the multiplicative harm downweight is the general case. Equation (12) quantifies the welfare cost of harm protection on the same yardstick used for g, so an operator can read it off the cost-quantification procedure of Section 7 without new machinery.

A soft weight does not guarantee protection; a hard constraint does. The multiplicative weight h is *soft*: a sufficiently large $\tau(x)V(x)$ can still clear the threshold despite a small $h(x)$, so h cannot guarantee any ceiling on the harm exposure of the treated set. When the operator needs a guarantee, the right object is a constraint, not a weight. Add to the budget a harm-exposure cap and solve

$$\max_{\pi \in \Pi_\kappa} \mathbb{E}\big[\pi(X_i)\, \tau(X_i) V_i g(R_i)\big] \quad \text{s.t.} \quad \mathbb{E}\big[\pi(X_i)\, \mathrm{harm}(X_i)\big] \le \bar{H}\, \kappa, \tag{13}$$

where $\mathrm{harm}(x)$ is a harm indicator and $\bar{H}$ caps the mean harm among the treated.

Corollary 2 (Harm-constrained welfare-optimal rule)**.** *Under Assumptions 1 and 2, the rule solving* (13) *ranks individuals by the harm-penalised score*

$$s^\ddagger(x) = \tau(x) V(x) g(R(x)) - \mu\, \mathrm{harm}(x), \tag{14}$$

treating those with $s^\ddagger(x) > \lambda$, where $\lambda \ge 0$ is the budget multiplier and $\mu \ge 0$ is the smallest value for which the harm constraint holds (with $\mu = 0$ when the constraint slacks). For any target harm level $\bar{H}$ achievable within the budget, this rule attains weakly higher welfare than any *rule meeting the same harm level, including a soft-weight rule h tuned to that level.*

Proof. Attach multipliers λ (budget) and $\mu \ge 0$ (harm) to the two constraints; the Lagrangian integrand is $\pi(x)\,[\tau V g - \mu\, \mathrm{harm} - \lambda]$, maximised pointwise by treating exactly when $s^\ddagger(x) = \tau V g - \mu\, \mathrm{harm} > \lambda$ (the same Neyman-Pearson argument as Proposition 1, now on the penalised score). Strong duality holds for the convex relaxation by the same Slater argument (the two-constraint feasible region is non-empty in its interior whenever a strictly feasible rule exists), and the relaxed optimum is again attained by a binary rule almost surely, because the boundary set $\{s^\ddagger(x) = \lambda\}$ is measure-zero under Assumption 2(i) applied to the penalised score. The dominance claim is immediate: the rule solves (13), so it is the welfare-maximiser over all feasible rules meeting the harm cap; a soft-weight rule achieving the same harm level is one such feasible rule, hence weakly worse. □

The practical reading: reserve the soft weight h for graded, sub-threshold harm signals where some trade-off against value is acceptable, and use the hard constraint (13) (or membership of the exclusion set $\mathcal{E}$, the $\bar{H} = 0$ limit) when a guarantee is required.

A worked harm-gate result. To make the trade-off concrete we add a harm covariate to the simulation, drawn so that harm rises with spend and therefore correlates positively with lifetime value (the configuration the paper warns about: high value can co-occur with high harm). Population mean harm is 0.42. We compare the unguarded value-weighted rule against the harm-protected rule that multiplies by a weight h falling from 1 to 0 as harm rises from 0.6 to 0.9, at $\kappa = 0.30$ over 200 replications. For the oracle rule (true components), harm protection reduces mean harm exposure among the treated from 0.466 to 0.411 (an 11.8% reduction) at a welfare cost of 6.4% (44,615 $\rightarrow$ 41,776): this is Corollary 1 made concrete, with the welfare cost read off the same yardstick as g. Confirming the warning at the head of this subsection, the unguarded score concentrates harm *above* the population mean (0.466 vs. 0.42), so without the gate value weighting does pull treatment toward higher-harm players. For the estimated rule the harm-exposure reduction is similar (0.475 $\rightarrow$ 0.411, about 13%) but the welfare *cost* is negative (16,824 $\rightarrow$ 19,400): in this DGP harm protection happens to remove high-value, low-effect players that the estimated score had over-ranked, so it improves operator welfare while reducing harm. That sign flip is the estimation-regime analogue of Remark 12 and should not be read as a general guarantee that harm protection is free; the oracle result is the reliable statement of its cost. Corollary 2 sharpens the design choice. Holding the harm-exposure level fixed at the 0.411 the soft gate achieves, the hard harm constraint of (13) attains welfare 43,552 against the soft gate's 41,776 (oracle, 200 replications), cutting the welfare cost of reaching that harm level from 2,839 to 1,063, a 62% reduction. The hard constraint sits on the welfare-harm efficient frontier that the soft multiplicative weight does not: when a guaranteed harm ceiling is the goal, the constraint is both safer (it cannot be overridden by a large τV) and cheaper in welfare than the soft weight.

Welfare-optimal targeting in the econometric sense is not the same as ethically optimal targeting in a harm-sensitive domain, and operators carry the responsibility for that distinction. We adopt the phrase *operator welfare* for the econometric quantity maximised by the composite score and reserve *player welfare* for the wellbeing of the individuals targeted; this paper's propositions are about operator welfare, while responsible-gaming safeguards protect player welfare, and the two can point in different directions.

6.3 Numerical Example

Example 1 (Illustrative targeting)**.** *Consider eight hypothetical players with known treatment effects, lifetime values, and churn risk scores (Table 2). The values in Table 2 are illustrative, chosen so that the three scoring rules produce visibly contrasting top-k sets. They do not attempt to respect the tight bound $\tau \leq p$ that a strictly binary-retention reading would impose (τ for a player with churn probability* 0.03 *is bounded above by* 0.03*). Readers should treat the numbers as abstract scores sufficient to illustrate ranking behaviour, not as a realistic simulation. We apply the composite score $s_i = \tau_i \cdot V_i \cdot g(p_i; \theta_0)$ with default parameters $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$ and a budget of* 3 *out of* 8 *players.*

The example illustrates three key effects:

***Pure uplift (τ alone)** selects players A, B, C. Player A has the highest treatment effect but very low churn risk ($p = 0.03$): they would retain without a bonus. The budget is wasted on a "sure thing."*

***Value-weighted uplift ($\tau \times V$)** selects C, B, D. Player A drops from first to fourth because their LTV is low (200 vs. 500–800 for B and C). This rule correctly prioritises high-value responders but treats all churn risk levels equally.*

***Full composite ($\tau \times V \times g$)** selects C, D, B. Player A is vetoed entirely ($g = 0$: churn too low to justify intervention). Player E is deprioritised ($g = 0.47$: high churn reduces the*

Table 2: Numerical example: eight hypothetical players. Columns show the treatment effect (τ, in percentage points; illustrative values, see the note preceding the table), lifetime value (V), churn risk score (p), the risk-modulation weight $g(p)$, the composite score s, and the targeting rank under three scoring rules. Budget: 3 players. The composite score s here multiplies τ in percentage points for readability, so the magnitudes are a factor of 100 larger than they would be if τ entered as a probability; ranks are unaffected. The welfare values reported in Section 7 use τ as a probability.

Player	τ (pp)	V	p	$g(p)$	s	Rank: τ	Rank: τV	Rank: $\tau V g$
A	+35	200	0.03	0.00	0	1	4	–
B	+30	500	0.25	0.57	8,571	2	2	3
C	+20	800	0.50	1.00	16,000	3	1	1
D	+15	600	0.45	1.00	9,000	4	3	2
E	+10	400	0.80	0.47	1,867	5	6	4
F	+5	1,200	0.10	0.14	857	7	5	5
G	−10	900	0.50	1.00	−9,000	8	8	8
H	+8	300	0.95	0.20	480	6	7	6

weight). Players C and D, both in the moderate-churn plateau ($g = 1$), dominate. The operator concentrates budget on the segment where interventions are judged most effective.

Player G is a "sleeping dog": an individual made worse off by treatment ($\tau < 0$). All three rules correctly give this player a negative score and exclude them from targeting. None of the three rules considers harm: a harm-sensitive operator would, in addition, apply the harm-protection weight h of Section 6.2 (and the exclusion set $\mathcal{E}$) before ranking, which can demote an otherwise high-scoring high-value player; the worked harm-gate result in Section 6.2 quantifies that effect.

7 Simulation Study

We evaluate the framework in a controlled simulation. The advantage of a simulation over a real data set is that we know the true treatment effects, true outcome values, and true risk-covariate values for every individual, so we can compare what each targeting rule achieves to the best-possible-given-the-truth benchmark. The trade-off is that the simulation's validity rests on our simulation world resembling a real targeting problem closely enough to be informative. We return to this point in Section 8.

The simulation is organised around three scenarios, each isolating when one component of the composite score adds value: (i) a well-specified *data-generating process* (DGP, the set of equations we use to generate the simulated individuals and outcomes) in which the treatment-effect estimator has access to all relevant features; (ii) a misspecified DGP in which the key feature driving the treatment-effect heterogeneity is hidden from the treatment-effect model; and (iii) a null DGP in which the treatment effect does not interact with the risk covariate at all. Section 7.2 presents the well-specified scenario. Sections 7.3 and 7.4 cover treatment-effect misspecification and the null DGP. Sections 7.5–7.7 report supporting analyses: budget sensitivity, the welfare-risk frontier, and robustness checks.

7.1 Design

We generate a simulated population of $N = 10{,}000$ individuals, each with four covariates. Treatment effects are nonlinear functions of the covariates, and the covariates also drive outcome value and churn probability, so the three components are correlated through the covariates as they are in practice. The full data-generating process follows.

Features. We generate four pre-treatment covariates independently of one another, one per individual:

$$\begin{aligned}
\text{account_age}_i &\sim \mathcal{N}(24, 8), \quad \text{clipped to } [1, 60], \\
\text{weekly_spend}_i &\sim \exp(\mathcal{N}(3.5, 0.8)), \quad \text{clipped to } [5, 500], \\
\text{login_freq}_i &\sim \text{Poisson}(4), \quad \text{clipped to } [1, 15], \\
\text{bonus_history}_i &\sim \mathcal{N}(3, 1.5), \quad \text{clipped to } [0, 10],
\end{aligned}$$

where $\mathcal{N}(\mu, \sigma)$ is the normal distribution, $\exp(\mathcal{N}(\cdot, \cdot))$ the log-normal (a right-skewed, strictly positive form for spend), and $\text{Poisson}(\lambda)$ the Poisson (integer login counts). "Clipped to $[a, b]$" means any value falling outside the range $[a, b]$ is pulled in to the nearest boundary, so the variable is guaranteed to live inside realistic bounds. This creates a small amount of probability mass at each endpoint; downstream estimators can in principle pick up on this concentration, though the effect is small at the clipping thresholds used here.

Let z_{spend}, z_{login}, z_{age}, z_{bonus} denote the *standardised* versions of each feature. Standardisation (mean-zero, unit-variance) makes the coefficients on the features in the equations below directly comparable. We compute the means and standard deviations once from a very large sample ($N = 2{,}000{,}000$ draws) rather than from each simulation replication's own data. This is a *held-out standardisation*: it prevents the training sample's noise from leaking into the test sample's coding of the features, a subtle source of bias when the standardisation is re-computed on each training set.

Churn probability. We generate each individual's churn probability from a *logistic model*, a standard way of producing a value between 0 and 1 as a function of covariates. The model first forms a linear combination of the covariates (called the *log-odds* or *logit*, because it equals $\log(p/(1-p))$ for a probability p):

$$\text{logit}(p_i) = -1.0 + 0.04\,(30 - \text{age}_i) - 0.25\,\text{login}_i + 0.01\,(40 - \text{spend}_i) + 0.1\,z_{\text{bonus},i} + \varepsilon_i^{(p)}, \quad (15)$$

where $\varepsilon_i^{(p)} \sim \mathcal{N}(0, 0.4)$ is an individual-level noise term. The scaling is deliberately mixed: age_i and spend_i enter as offsets from constants (30 and 40) so that the intercept reflects churn at reference values for those features rather than at zero; login_i enters in raw units, appropriate for a count with a natural scale; and bonus_history_i enters standardised so that its coefficient is unit-free. The offsets (30, 40) are pedagogical anchors, not the feature means; they shift the intercept but leave the covariance structure unchanged. We convert the log-odds to a probability with the sigmoid $\text{sigmoid}(z) = 1/(1+e^{-z})$: $p_i = \text{sigmoid}(\text{logit}(p_i))$. We then clip p_i to $[0.01, 0.99]$ to avoid degenerate extremes.

Lifetime value.

$$V_i = 10\,\text{spend}_i + 40\,\text{login}_i + 5\,\text{age}_i + \varepsilon_i^{(V)}, \quad \varepsilon_i^{(V)} \sim \mathcal{N}(0, 80), \quad \text{clipped to } [50, 8000]. \quad (16)$$

CATE with churn moderation. The treatment effect features an inverse-U interaction with churn probability:

$$\begin{aligned}
\tau_{\text{base},i} &= 0.06 + 0.10\,z_{\text{login},i} - 0.02\,z_{\text{login},i}^2 - 0.04\,z_{\text{spend},i}, \\
m(p_i) &= 4\,p_i\,(1 - p_i), \\
\tau_i &= \tau_{\text{base},i} \cdot m(p_i) + \varepsilon_i^{(\tau)}, \quad \varepsilon_i^{(\tau)} \sim \mathcal{N}(0, 0.03), \quad \text{clipped to } [-0.20, 0.40].
\end{aligned} \quad (17)$$

The moderation function $m(p) = 4p(1-p)$ peaks at $p = 0.5$ and equals zero at $p \in \{0, 1\}$, creating the structural feature that risk modulation is designed to exploit: treatment effects are highest at moderate churn and attenuated at both extremes. The symmetric shape is a

simulation convenience: Ascarza (2018) documents asymmetric attenuation in real retention settings (treatment effects tend to fall off more sharply at high churn than at low churn), and an operator using this framework on real data should not assume the moderation is symmetric.

Baseline and outcomes. Each individual has a baseline retention probability (the probability they would remain active if not treated), and outcomes are Bernoulli draws:

$$Y_i(0) \sim \text{Bernoulli}\big((0.6 + 0.15\, z_{\text{login},i} + 0.05\, z_{\text{spend},i} - 0.08\, z_{\text{age},i} + \varepsilon_i^{(b)})_{[0.05,\,0.95]}\big), \tag{18}$$

with $\varepsilon_i^{(b)} \sim \mathcal{N}(0, 0.05)$. Let baseline_i denote the clipped linear predictor (the expression inside the Bernoulli). The potential outcome under treatment is $Y_i(1) \sim \text{Bernoulli}((\text{baseline}_i + \tau_i)_{[0.01,\,0.99]})$: treatment simply shifts the retention probability by the treatment effect τ_i. For individuals with baseline retention close to 1 and positive τ_i, the clip to $[0.01, 0.99]$ can activate, so the *effective* treatment-effect realised in the outcome is smaller than the nominal τ_i. Welfare on the test set is evaluated using nominal $\tau_i \cdot V_i$ (the DGP's generated treatment effect), not the clip-adjusted effective value, so reported welfare numbers overstate the realised lift for high-baseline individuals. The bias is not uniform: strategies that concentrate on the high-V tail (value-weighted and joint-model) are flattered most, so it points toward inflating the very comparisons we foreground. A clip-adjusted re-evaluation (Section 7.7) confirms the bias is modest (3.7–6.7%) and changes no ranking. We assign treatment with a fair coin flip, $W_i \sim \text{Bernoulli}(0.5)$, mimicking a randomised experiment. This design trivially satisfies the overlap condition of Assumption 1(ii) with $\eta = 0.5$ and does not stress-test the low-propensity regime typical of observational deployments, where $P(W = 1 \mid X)$ can sit well below 0.5; robustness under non-trivial propensity variation is outside the scope of the present simulations.

Estimation protocol. We run 200 independent replications of the following protocol to characterise variability across simulation runs. Each replication $r \in \{0, \ldots, 199\}$ uses the fixed seed $42 + 7r$, so the entire study is deterministic and reproducible; the simulation code, the revision-additions code (asymmetric DGP, harm gate, effective-welfare, and the S-Learner under misspecification and the null DGP), and the resulting output files are released in the companion repository.

Train/test split. We partition the population 70/30 into a training set (used to fit all models) and a test set (used only for final evaluation). This prevents each strategy from inadvertently profiting from noise in the data it was trained on, a practice known in ML as out-of-sample evaluation.

Component models. We estimate three models on the training data: a *T-Learner* for the treatment effect, which fits one outcome model on the treated group and one on the control group using gradient boosted trees (an ensemble ML method that combines many shallow decision trees, each correcting the errors of the previous ones) and takes the difference; a gradient boosted regression model for outcome value; and a gradient boosted classifier for churn probability (a similar ensemble, predicting a probability instead of a continuous value).

Value-model misspecification. We deliberately misspecify the value model: the training target uses the form $12 \cdot \text{spend}^{0.9} + 35 \cdot \text{login} + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 200)$, which drops the age feature and uses a nonlinear transformation of spend. This mimics the realistic situation where the true value function is unknown and the analyst has to pick a plausible but wrong functional form.

Churn model. The churn model is trained on the control-group non-retention indicator, $1 - Y_i \mid W_i = 0$ (one minus the retention outcome for individuals who did not receive the bonus). This is a reasonable proxy for churn risk but is not identical to the structural churn probability used in the DGP; see Remark 13.

Evaluation. All component models are estimated on training data alone. The true ("oracle") values of τ, V, and R, known to us because we generated them, are used only to compute

welfare on the test set and to serve as a theoretical upper bound on what any estimated rule could achieve. We report two reference budgets rather than one. The realistic operating budget is $\kappa = 0.10$ (treat the top 10% of the test population), which matches real marketing-operations practice, where a single retention campaign typically reaches 5% to 15% of the active base; this is the budget on which an operator's deployment decision actually turns. We also report $\kappa = 0.30$ as a diagnostic budget: it gives every strategy enough latitude to express its preferences over the population, which makes the mechanism behind each result easier to see, but it is more generous than most campaigns. The two budgets tell different stories, and the difference is itself a finding (Section 7.5): Section 7.5 varies κ across $\{0.10, 0.20, 0.30, 0.50\}$, and the ranking of strategies depends on κ.

Robustness. As an additional check we re-estimate the treatment effect using an *X-Learner* (Künzel et al., 2019), a metalearner that first imputes individual-level treatment-effect estimates by combining each unit's observed outcome with a model-predicted counterfactual, then fits separate treatment-effect models on the treated and control groups; results appear in Section 7.7.

Calibrated parameter selection. The preference-weight parameters θ are picked by held-out grid search: on each replication we further split the training set 60/40 into an *estimation fold* (used to fit the three component models) and a *validation fold* (used only to score the candidate θ values). We test 22 parameter configurations, including a near-identity configuration with $p_L = 0$, $p_{ML} = 0.01$, $p_{MH} = 0.99$, $p_H = 1$, $\phi = 1$. Under this parameterisation $g(p) = 1$ everywhere except on a vanishingly small ramp $p \in [0, 0.01)$, so it effectively turns g into the constant function 1 and lets calibration decline any preference weighting if it does not help. For each candidate, we compute the composite score on the validation fold using the estimated components $(\hat{\tau}, \hat{V}, \hat{R})$, apply the resulting rule, and measure its welfare. We pick the configuration with the highest validation-fold welfare. No oracle information enters parameter selection.

Statistical testing. We compare strategies with paired t-tests across the 200 replications, applying a Holm-Bonferroni correction within each scenario's family of comparisons (well-specified, misspecified, null DGP, budget sensitivity, robustness); we do not correct across families, treating each as a distinct question. We report Cohen's d alongside every p-value, because the paired design makes even small mean differences statistically significant. The precision yardstick is the standard deviation of the paired *differences* (of order 850–900 units for the headline comparisons), so with 200 replications the minimum difference detectable at $p < 0.05$ is roughly 120–125 units, about 0.7% of the $\kappa = 0.30$ value-weighted welfare: statistically resolvable but operationally small, and we do not treat such gaps as large. A Wilcoxon signed-rank check (not reported) leaves every conclusion unchanged.

Strategies evaluated. We compare seven targeting strategies (the results tables add an eighth row, a τV oracle that ranks by the true $\tau \cdot V$ with $g \equiv 1$: the welfare-maximising rule under $\mathcal{W}(\pi; \mathbf{1})$ and hence the highest-welfare point achievable in this DGP): (i) *Random*: uniform random selection, providing a lower benchmark; (ii) *CATE Only*: rank by the estimated treatment effect $\hat{\tau}(x)$ alone; (iii) *CATE* $\times$ *LTV*: rank by the product of estimated treatment effect and estimated outcome value, $\hat{\tau}(x) \cdot \hat{V}(x)$, the current practitioner heuristic; (iv) *Full Framework (Est.)*: rank by $\hat{\tau}(x) \cdot \hat{V}(x) \cdot g(\hat{R}(x); \theta_0)$ with default (uncalibrated) preference parameters $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$; (v) *Full Framework (Cal.)*: same as (iv) but with the preference parameters chosen by the held-out calibration procedure; (vi) *Full Framework (Oracle)*: the estimation-free benchmark for the decomposed framework at θ_0, ranking by the true (not estimated) components $\tau(x) \cdot V(x) \cdot g(R(x); \theta_0)$; because $g(\cdot; \theta_0) \not\equiv 1$, this is not the welfare-maximising rule under $\mathcal{W}(\pi; \mathbf{1})$ (by Proposition 3), but it bounds how much of the framework's welfare gap to the estimated versions is attributable to estimation error in $\hat{\tau}, \hat{V}, \hat{R}$; (vii) *S-Learner*: a single gradient boosted outcome model that takes treatment as an input feature

(max tree depth 4, compared with 3 for the T-Learner), producing a treatment-effect estimate by subtracting its prediction at $W = 0$ from its prediction at $W = 1$: $\hat{\tau}_{\text{SL}}(x) = \hat{f}(x, 1) - \hat{f}(x, 0)$, then ranking by $\hat{\tau}_{\text{SL}}(x) \cdot \hat{V}(x)$ (Künzel et al., 2019).

The S-Learner is a single-model policy alternative; it does not use the three-component decomposition for the treatment-effect estimate, but it still borrows $\hat{V}$ from the decomposed pipeline (a fully end-to-end alternative would predict the value-weighted outcome directly). In the taxonomy we use from here on, this borrowed-$\hat{V}$ S-Learner is the *hybrid* strategy (a joint treatment-effect model with a separate value model), as distinct from the *decomposed* rule ($\hat{\tau} \cdot \hat{V}$ from two separate models) and the *end-to-end* rule (a single model for the value-weighted outcome, Section 7.7); Section 8 shows the hybrid is the highest-welfare estimated strategy of the three. The deeper trees give the S-Learner more room to capture treatment-covariate interactions. Section 7.7 reports a depth-matched ablation (S-Learner at depth 3; T-Learner at depth 4 paired with the existing LTV model) which shows the joint-modelling advantage persists, and modestly widens, once the tree-depth hyperparameter is equalised in either direction.

7.2 Main Results

Table 3 reports the seven strategies side by side with the τV-oracle benchmark (eight rows in total); Figure 2 plots the welfare column visually. All welfare figures use the oracle welfare function $\mathcal{W}(\pi; \mathbf{1}) = \sum_{i \in \text{targeted}} \tau_i \cdot V_i$ on the test set, giving every strategy a common yardstick. This choice has a consequence worth flagging: by Proposition 3, any strategy that optimises a surrogate with $g \neq 1$ weakly underperforms the $g \equiv 1$-optimal strategy under this yardstick. That is by design; an operator who chooses $g \neq 1$ is expressing a targeting preference and accepting the welfare cost the proposition quantifies.

Table 3: Targeting strategy comparison (N = 10,000, 200 replications, $\kappa = 0.30$). ATE is the average treatment effect among treated (percentage points). Welfare throughout is *operator* welfare, total value generated under $\mathcal{W}(\pi; \mathbf{1}) = \sum \tau_i \cdot V_i$ on the test set (every "welfare" figure in the paper is operator welfare in the sense of Section 6.2, not player wellbeing). Risk exposure is mean churn probability among treated. Churn Mod is the mean churn moderation $4p(1 - p)$ among treated. Values shown as ± are standard deviations across replications; pairwise significance tests are in Table 4.

Strategy	ATE (pp)	Welfare	Risk Exp.	Churn Mod
Random	1.2 ± 2.1	$5{,}399 \pm 1{,}359$	0.166	0.512
CATE Only	3.3 ± 1.9	$16{,}634 \pm 2{,}954$	0.153	0.485
CATE × LTV	3.0 ± 2.0	$16{,}427 \pm 3{,}090$	0.136	0.440
Full Framework (Est.)	2.8 ± 2.0	$13{,}629 \pm 3{,}008$	0.146	0.468
Full Framework (Cal.)	3.0 ± 2.0	$16{,}427 \pm 3{,}090$	0.136	0.440
Full Framework (Oracle)	6.7 ± 2.0	$36{,}641 \pm 704$	0.192	0.596
τV Oracle ($g \equiv 1$)	6.8 ± 1.8	$44{,}615 \pm 805$	0.138	0.451
S-Learner	3.9 ± 2.0	$24{,}310 \pm 4{,}098$	0.120	0.399

Four patterns stand out. First, CATE-based targeting triples welfare relative to random allocation (16,634 vs. 5,399), confirming that heterogeneity in treatment effects is worth exploiting. Second, the calibration procedure selects the near-identity configuration in 197 of 200 replications, so the calibrated strategy reproduces CATE × LTV (16,427) almost exactly. This is the behaviour that Proposition 3 predicts for the oracle rule, and the calibration procedure tracks it empirically because estimation is accurate enough in this DGP: the calibration objective is estimated welfare under $g \equiv 1$, which Proposition 3 identifies as the welfare-maximising

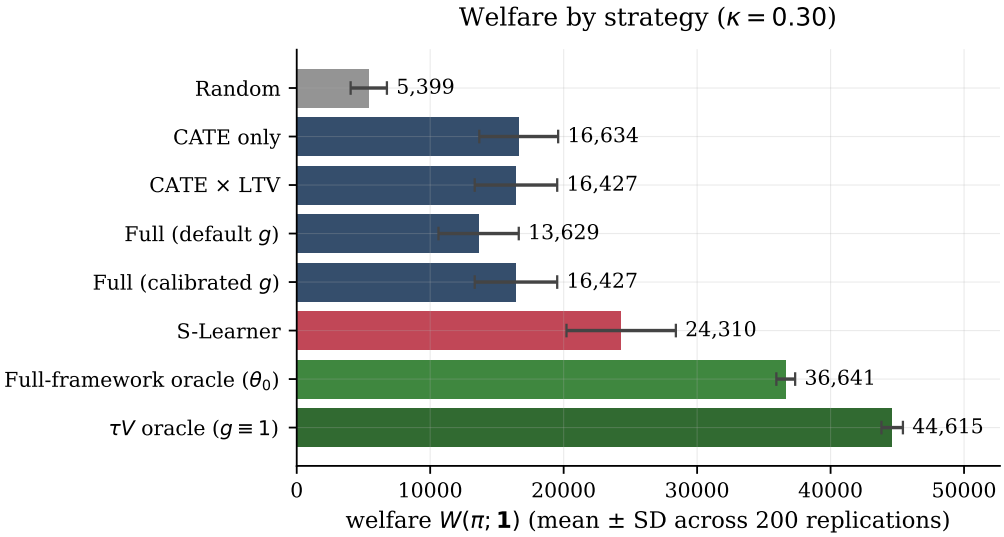


Figure 2: Welfare by targeting strategy ($\kappa = 0.30$, 200 replications). Error bars are ± 1 standard deviation. The S-Learner (red) beats every decomposed estimated strategy; the Full-framework oracle (light green) is the estimation-free benchmark for the decomposed framework at θ_0; the τV oracle (dark green) is the welfare-maximising rule under $\mathcal{W}(\pi; \mathbf{1})$ ($g \equiv 1$) and the highest-welfare point achievable in this DGP.

Table 4: Pairwise welfare differences (Holm-Bonferroni corrected, 200 replications). Positive values favour the first strategy.

Comparison	Diff.	t	p_{holm}	Note
CATE Only vs. Random	+11,235	50.61	< 0.0001	–
CATE × LTV vs. Random	+11,029	48.38	< 0.0001	–
Cal. vs. CATE × LTV	−1	−0.97	0.333	calibration picks $g \approx 1$
Cal. vs. CATE Only	−207	−3.27	0.0026	$d = -0.231$
Default Est. vs. CATE × LTV	−2,799	−45.59	< 0.0001	default g costs welfare
Oracle vs. CATE × LTV	+20,214	92.74	< 0.0001	estimation is the bottleneck

choice, so calibration approaches the identity as estimates approach truth. The calibration procedure's substantive output is the frontier of welfare costs across configurations (Section 7.6). Third, the default (uncalibrated) g costs 17% of welfare (13,629 vs. 16,427), making the value-conservatism trade-off quantitative at one representative parameterisation. By Proposition 3, any fixed-parameter g other than the identity imposes some welfare cost; 17% is simply the cost at the pedagogical θ_0, and a different θ would produce a different cost that the frontier analysis in Section 7.6 maps out. Note that the calibrated framework slightly underperforms pure CATE targeting at this budget: 16,427 vs. 16,634, a gap of 207 welfare units ($d = -0.231$, $p_{\text{holm}} = 0.0026$). This gap is statistically resolvable (it exceeds the ~ 125-unit detection floor derived above, consistent with $p_{\text{holm}} = 0.0026$) but operationally small: 207 units is about 1.2% of welfare, and the effect size is small ($d = -0.231$). The magnitude matters less than the fact that its *sign* reverses with budget: at tighter budgets CATE Only underperforms CATE × LTV, and at looser budgets it overperforms (Section 7.5). That reversal, not the size of any single gap, is the substantive point. Fourth, the S-Learner achieves 24,310, substantially above every decomposed estimated strategy. The Full-framework oracle at θ_0 scores 36,641, and the τV oracle (the welfare-maximal rule under $\mathcal{W}(\pi; \mathbf{1})$, i.e. $g \equiv 1$) scores 44,615. The gap between these two (7,974 welfare units) is the cost of the default θ_0 preference alone, independent of any estimation error; the gap between the S-Learner and the τV oracle (20,305 units) is the estimation-error cost under the common $\mathcal{W}(\pi; \mathbf{1})$ yardstick. The S-Learner gap is the welfare-interpretability trade-off; we return to it in Section 8.

7.3 CATE Misspecification

Design. The CATE model is re-estimated with login frequency dropped from its feature set; LTV and churn models retain all four features. This simulates a realistic scenario where the uplift model is imperfect while auxiliary models are better specified. All other details are identical to the primary simulation.

Table 5: CATE misspecification results: the CATE model drops login frequency; LTV and churn models retain all features. Same DGP, $N = 10{,}000$, 200 replications, $\kappa = 0.30$. p-values are Holm-Bonferroni corrected. Risk exposure and churn moderation columns are omitted as they are qualitatively similar to the well-specified scenario.

Strategy	ATE (pp)	Welfare
CATE Only	1.1	6,700 ± 2,056
CATE × LTV	1.4	6,955 ± 2,337
Full Framework (Est.)	0.9	3,809 ± 2,339
Full Framework (Cal.)	1.4	6,954 ± 2,336
S-Learner	3.9	24,310 ± 4,098
Cal. vs CATE Only: +254, $p < 0.0001$, $d = 0.303$ *(Cal. significantly better)*		
Cal. vs CATE × LTV: −1, $p = 0.42$, $d = -0.057$ *(Cal. = CATE × LTV, $g \approx 1$ selected)*		

All strategies degrade. CATE Only drops from 16,634 to 6,700 (a 60% welfare loss), confirming login frequency as the primary driver. The calibrated framework (6,954) outperforms CATE Only (+254, $p < 0.0001$, $d = 0.303$), and it matches CATE × LTV (6,954 vs. 6,955; not significant) because calibration again picks $g \approx 1$. The +254 gap is statistically resolvable (well above the ~ 125-unit detection floor of Section 7.1, with $d = 0.303$), though still operationally modest at roughly 3.8% of the misspecified-scenario welfare; we read it as a real but small improvement. So the improvement over CATE Only comes from value weighting, not from g. When the CATE model is degraded, the LTV model's features give $\hat{\tau} \cdot \hat{V}$ information the CATE estimator alone cannot reach; this is the mechanism described in Remark 12, operating

in the limit where LTV alone absorbs the missing moderation structure. The S-Learner appears unaffected by this misspecification: its welfare (24,310) is identical to the well-specified scenario, because the login-frequency dropout is applied only to the decomposed T-Learner CATE, while the joint model still sees all four features. That comparison is not like-for-like, and reading it as "the joint model is more robust" would be a mistake. When we drop login from the S-Learner's feature set as well, it degrades by 63.8% (welfare 24,310 $\rightarrow$ 8,803, $p < 0.001$, $d = -2.94$), a proportional loss comparable to the decomposition's 60%. So the joint model is not more robust to feature dropout; both architectures are fragile to losing an informative feature, and the decomposition's apparent fragility here is really the artefact of dropping the feature from only one side of the comparison. What the joint model buys, given the *same* feature set, is an architectural advantage (it exploits cross-component interactions the decomposition cannot), not robustness to missing inputs.

An adversarial misspecification: when value weighting cannot rescue. The result above is favourable to value weighting by construction: the dropped feature (login frequency) enters lifetime value linearly (Equation 16), so the value model recovers the signal the CATE model lost. That is a best case, not a general one. To bound it we run the adversarial counterpart: a DGP in which the treatment effect also depends on *bonus history* (a feature *absent* from the value equation), and a CATE model that drops bonus history. The value model now cannot recover the lost signal, because bonus history never enters V. At $\kappa = 0.30$ value weighting no longer helps (CATE Only 18,190 vs. CATE $\times$ LTV 18,088; difference -101, 0.6%, $p = 0.23$, not significant), and at the tight budget $\kappa = 0.10$ it actively hurts (CATE Only 5,186 vs. CATE $\times$ LTV 3,862; difference $-1{,}324$, 25.5%, $p < 0.001$). Contrast this with the favourable login-drop case, where value weighting *beat* pure CATE (6,955 vs. 6,700). The claim that value weighting helps most when the treatment-effect model is weakest therefore holds only when the value model shares information with the dropped signal; when the misspecification removes a driver orthogonal to value, value weighting cannot compensate and, at tight budgets, makes matters worse. The S-Learner, which here retains all features including bonus history, again scores highest (41,416 at $\kappa = 0.30$); but, as in the login-drop case, this reflects its larger information set rather than robustness, since dropping bonus history from the S-Learner too degrades it by 36% (41,416 $\rightarrow$ 26,486, $p < 0.001$).

7.4 Null DGP: No CATE-Risk Interaction

We repeat the simulation with $m(p) \equiv 1$ (no CATE-churn interaction), so that $\tau_i = \tau_{\text{base},i} + \varepsilon_i^{(\tau)}$. All other aspects remain identical.

Table 6: Null DGP results: no CATE-churn interaction ($m(p) \equiv 1$). Same estimation protocol, $N = 10{,}000$, 200 replications, $\kappa = 0.30$.

Strategy	Welfare
CATE Only	$63{,}098 \pm 5{,}549$
CATE $\times$ LTV	$68{,}256 \pm 6{,}324$
Full Framework (Est.)	$53{,}887 \pm 6{,}484$
Full Framework (Cal.)	$68{,}256 \pm 6{,}324$
S-Learner	$86{,}347 \pm 5{,}352$
Cal. vs CATE $\times$ LTV: diff = 0, Cal. = CATE $\times$ LTV ($g \approx 1$ selected)	

Under the null DGP, CATE $\times$ LTV (68,256) outperforms CATE Only (63,098), and the calibrated framework reproduces CATE $\times$ LTV because the calibration procedure selects the near-identity configuration as it must (again by Proposition 3). The substantive empirical content

of this scenario is that value weighting beats pure CATE targeting even when there is no risk-moderation structure in the DGP to exploit. The S-Learner again leads (86,347), consistent with its dominance in the main and misspecified scenarios.

7.5 Budget Sensitivity

Table 7 reports welfare across four budget levels.

Table 7: Budget sensitivity: welfare across budget fractions ($N = 10{,}000$, 200 replications).

Strategy	$\kappa = 0.10$	$\kappa = 0.20$	$\kappa = 0.30$	$\kappa = 0.50$
CATE Only	4,766	10,750	16,634	26,740
CATE × LTV	3,268	9,885	16,427	26,941
Full Framework (Est.)	2,586	7,960	13,629	25,171
Full Framework (Cal.)	3,268	9,884	16,427	26,941
S-Learner	7,563	16,301	24,310	36,033
τV Oracle ($g \equiv 1$)	21,281	34,765	44,615	56,145

At every budget level the calibrated framework matches CATE × LTV within noise (paired t-tests, all $p > 0.05$). The more interesting pattern, and the one the paper's title claim rests on, is that the ranking between value-weighted targeting and CATE Only depends on budget. Because this comparison carries a headline claim, we give it the same inferential treatment as the main table rather than reporting means alone. Table 8 reports the paired difference (CATE × LTV minus CATE Only) at each budget, Holm-Bonferroni corrected across the four-budget family, with Cohen's d and a 95% confidence interval. The difference is negative and large at $\kappa = 0.10$ (CATE Only ahead by 1,499 welfare units, $d = -1.44$), shrinks through $\kappa = 0.20$ and 0.30, and turns positive at $\kappa = 0.50$ (value weighting ahead by 201, $d = 0.45$); every difference is resolvable after correction ($p_{\text{holm}} \leq 0.0014$).

Table 8: Budget-sensitivity inference: paired difference CATE × LTV minus CATE Only at each budget ($N = 10{,}000$, 200 replications; negative favours CATE Only). Holm-Bonferroni corrected across the four budgets. The crossover budget $\kappa^\star$ (where the difference changes sign) is estimated by linear interpolation of the mean-difference curve on a grid $\kappa \in \{0.05, 0.075, \ldots, 0.50\}$, with a 95% CI from 2,000 replication-level bootstrap resamples.

	$\kappa = 0.10$	$\kappa = 0.20$	$\kappa = 0.30$	$\kappa = 0.50$
Diff. (×LTV − Only)	−1,499	−865	−206	+201
Cohen's d	−1.44	−0.77	−0.23	+0.45
p_{holm}	< 0.001	< 0.001	0.0014	< 0.001
95% CI	$[-1{,}644, -1{,}354]$; $[-1{,}021, -709]$; $[-331, -81]$; $[138, 264]$			

The crossover budget is $\kappa^\star = 0.34$ (95% bootstrap CI $[0.32, 0.37]$) under the symmetric main moderation, and earlier, at $\kappa^\star = 0.25$ (95% CI $[0.22, 0.28]$), under the asymmetric moderation of Section 7.7: value weighting begins to pay at a lower budget when treatment-effect attenuation is asymmetric. In both cases the crossover is estimated directly from the welfare curves, not from the heuristic variance argument below. The practical recommendation to use $\hat{\tau} \cdot \hat{V}$ therefore applies most strongly at moderate-to-generous budgets (κ above roughly 0.3). At very tight budgets, pure CATE targeting may be preferable because value weighting dilutes the uplift signal by prioritising high-LTV individuals whose treatment effects may be modest.

A short heuristic argument underlies the reversal. It is heuristic, not exact: it assumes the estimation errors are mean-zero and independent of (τ, V), whereas T-Learner errors are neither (they correlate with baseline outcome levels, which share covariates with V), so the following should be read as indicating the mechanism rather than deriving the crossover. Write the estimated components as $\hat{\tau} = \tau + \epsilon_\tau$ and $\hat{V} = V + \epsilon_V$ with mean-zero errors independent of τ, V; then

$$\hat{\tau}\hat{V} \;=\; \tau V \;+\; V\,\epsilon_\tau \;+\; \tau\,\epsilon_V \;+\; \epsilon_\tau\,\epsilon_V,$$

so the conditional variance of the product-score at a given (τ, V) is approximately $V^2\,\mathrm{Var}(\epsilon_\tau) + \tau^2\,\mathrm{Var}(\epsilon_V)$ (dropping the second-order term). The V^2 factor amplifies $\hat{\tau}$-noise most strongly in the right tail of V. Ranking by $\hat{\tau}$ alone escapes this amplification. At tight κ, the top-κ set is drawn entirely from the joint right tail where the amplification is largest, so the product estimator misranks more often than $\hat{\tau}$ alone and the signal gained from including $\hat{V}$ does not offset the extra variance. As κ grows, the top-κ set extends into the body of V where amplification is milder, and the signal carried by $\hat{V}$ (moderate-τ individuals with very high V) dominates; the product estimator then overtakes $\hat{\tau}$. The crossover $\kappa^\star$ depends on $\mathrm{Var}(\epsilon_\tau), \mathrm{Var}(\epsilon_V)$, and the joint tail of (τ, V), and is therefore DGP-specific; we estimate it directly (with a bootstrap confidence interval) rather than from this argument in Section 7.5.

The S-Learner's welfare advantage over the decomposed strategies, first reported at $\kappa = 0.30$ in Table 3, persists across budgets: it is the highest-welfare estimated strategy at every κ.

7.6 Value-Conservatism Frontier

Table 9 and Figure 3 report the welfare-risk trade-off across six configurations of g. All welfare figures use $\mathcal{W}(\pi; \mathbf{1}) = \sum \tau_i \cdot V_i$ on the test set.

Table 9: Welfare and risk exposure under different g configurations ($N = 10{,}000$, 200 replications, $\kappa = 0.30$). Welfare evaluated under $g \equiv 1$ for every configuration. Labels show $(p_L/p_{ML}/p_{MH})$; all configurations use $p_H = 0.90$ and $\phi = 0.20$ unless stated.

Configuration	Welfare	Risk Exp.	Churn Mod
No risk modulation ($g \equiv 1$)	16,427	0.136	0.440
Wide (0.03/0.50/0.70)	12,588	0.148	0.472
Default (0.05/0.40/0.60)	13,629	0.146	0.468
Narrow (0.10/0.30/0.50)	14,940	0.145	0.464
Aggressive (0.15/0.35/0.50/ϕ=0.05)	13,713	0.149	0.476
Calibrated	16,427	0.136	0.440

No fixed-parameter configuration exceeds the unmodulated welfare of 16,427, and the calibrated configuration matches it exactly by picking $g \approx 1$. One caveat on the frontier itself: in this DGP, risk exposure only varies between 0.136 and 0.149 across configurations, a range of roughly 10%, while welfare varies from 12,588 to 16,427 (about 25%). Changes in g thus move welfare considerably more than they move risk exposure in this simulation. The operator's dial on g is therefore mostly a welfare cost dial, not a dramatic reshaping of who gets treated; in settings where the risk covariate is more concentrated in the treated tail (for example, a population where churn risk clusters sharply), the frontier would show wider risk-exposure variation.

7.7 Robustness

Misspecification of g. We test sensitivity by inverting the risk-modulation function so that it puts high weight at extreme churn levels and low weight in the moderate zone (the opposite of the

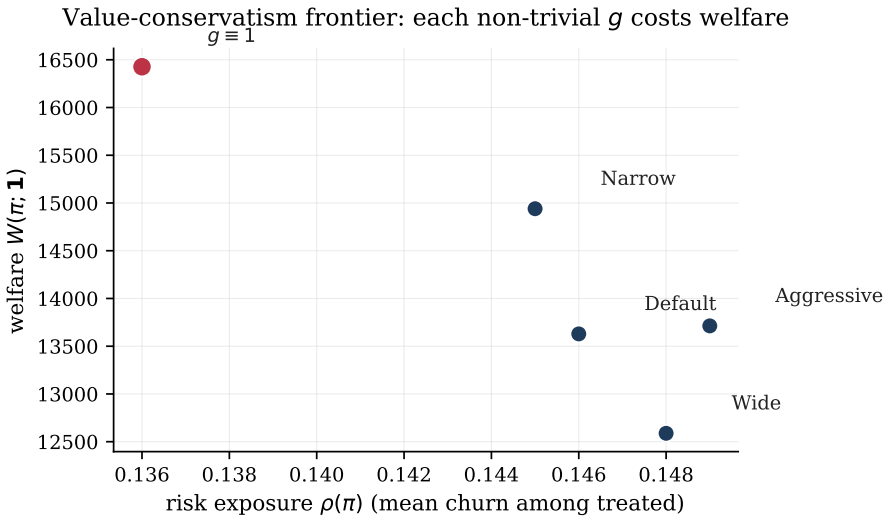


Figure 3: Value-conservatism frontier: each fixed-parameter g sits below the unmodulated $g \equiv 1$ baseline in welfare. The calibrated configuration overlaps the baseline. This is Proposition 3 made visual.

default). The inverted g achieves welfare 13,905 with risk exposure 0.145, versus 13,629 for the default and 16,427 for $g \equiv 1$. Both non-identity configurations underperform the unmodulated baseline, consistent with Proposition 3.

Depth-matched S-Learner. The published S-Learner uses a deeper tree ensemble (max depth 4) than the decomposed T-Learner (max depth 3), so the welfare comparison reported in Table 3 mixes joint modelling with a hyperparameter difference. We re-run two depth-matched variants on the same 200 seeds: (i) an S-Learner at max depth 3 and (ii) a T-Learner at max depth 4 paired with the existing LTV model. The S-Learner at depth 3 achieves welfare $25{,}213 \pm 4{,}522$ (slightly higher than the published depth-4 S-Learner at 24,310), while the T-Learner at depth 4 paired with LTV achieves $14{,}000 \pm 2{,}738$ (lower than the published depth-3 T-Learner × LTV at 16,427, indicating modest overfitting at the deeper capacity). At common depth 3 the joint-versus-decomposed gap is thus +8,786 welfare units (+53%); at common depth 4 it is +10,310 (+74%). The welfare advantage of joint modelling over the T-Learner-based decomposition in this DGP therefore persists, and modestly widens, when the tree-depth hyperparameter is equalised in either direction.

Asymmetric-moderation DGP. The main DGP uses a symmetric moderation $m(p) = 4p(1-p)$, which is the most favourable structure for the risk-modulation mechanism and, as Ascarza (2018) notes, unrealistic: real attenuation falls off faster at high churn than at low churn. We therefore re-run the simulation with an asymmetric moderation $m(p) \propto p(1-p)^3$ (peaking near $p = 0.25$ and decaying sharply toward high churn), a structurally different second DGP. The qualitative pattern is unchanged. At $\kappa = 0.30$: CATE Only $36{,}444 \pm 3{,}727$, CATE × LTV $36{,}861 \pm 4{,}322$, Full Framework (Cal.) $36{,}861 \pm 4{,}322$, Full Framework (Est.) $30{,}616 \pm 4{,}297$, Full Framework (Oracle) $54{,}660 \pm 964$, and S-Learner $49{,}991 \pm 4{,}080$. Calibration again selects the near-identity configuration (198 of 200 replications), so Cal. reproduces CATE × LTV; the default fixed g again costs welfare; and the S-Learner again dominates every decomposed estimated strategy. Value weighting now slightly helps rather than slightly hurts at this budget (36,861 vs. 36,444), but the three load-bearing findings (calibration selects the identity g, joint modelling wins, value weighting's sign is budget- and structure-dependent) are robust to the moderation structure. This sign change is itself a deployment hazard worth stating plainly: whether value weighting helps or hurts at a given budget depends on the moderation shape,

which the operator cannot observe ex ante. The safe reading is that the value-weighting decision should be made empirically per deployment (via held-out or off-policy evaluation), not assumed from this paper's main-DGP sign.

End-to-end joint model (un-handicapped comparator). The S-Learner in Table 3 borrows the decomposed pipeline's value estimate $\hat{V}$, so it is not a fully joint comparator. We therefore also run the end-to-end alternative: a single gradient-boosted model trained to predict the value-weighted outcome ($\hat{V}_i^{\text{proxy}} \cdot Y_i$) directly, scored by the treated-minus-control difference, with no separate value model. It achieves welfare $20{,}158 \pm 6{,}194$, above the best decomposed estimated strategy (CATE $\times$ LTV, 16,427) but below the borrowed-$\hat{V}$ S-Learner (24,310). Two things follow. First, joint modelling beats the decomposition even when it does not borrow the value estimate, so the welfare advantage is a property of joint modelling rather than an artefact of the comparison. Second, the borrowed-$\hat{V}$ S-Learner's extra margin over the end-to-end model shows that a well-specified separate value estimate is itself worth something; the strongest estimated strategy in this DGP combines a joint treatment-effect model with a separate value model, which is neither the pure decomposition nor the pure end-to-end joint model.

Diagnostic decomposition (does it finger the right component?). The regret decomposition (Remark 9) claims to identify which component's estimation error drives misranking. We test this directly by computing its two terms on the misranked set $\mathcal{M}$ (the symmetric difference of the oracle and estimated top-κ sets) at $\kappa = 0.30$. Under the well-specified pipeline the CATE-error term $|\tau - \hat{\tau}| \cdot |V|$ averages 81 welfare units on $\mathcal{M}$ against 21 for the value-error term $|\hat{\tau}| \cdot |V - \hat{V}|$, so roughly 80% of near-threshold score error is attributable to treatment-effect estimation, correctly flagging it as the binding component (consistent with the large oracle-to-estimated welfare gap in Table 3). When the CATE model is deliberately degraded (login dropped), the CATE-error term rises to 90 while the value-error term is unchanged at 21: the decomposition localises the added error to exactly the component that was degraded, and leaves the other untouched. This converts the diagnostic from an asserted property into a demonstrated one.

Heavy-tailed value. Gaming lifetime value is extreme-tailed. Replacing the additive value with a multiplicative log-normal tail (orders-of-magnitude spread, clipped at 100,000) leaves the pattern intact and, if anything, slightly sharpens it: at $\kappa = 0.30$ value weighting underperforms pure CATE by 2.9% (CATE Only 27,360 vs. CATE $\times$ LTV 26,559), consistent with the variance-amplification argument of Section 7.5 (a heavier V tail amplifies $\hat{\tau}$-noise in the product more), and the S-Learner again leads (40,527).

Further checks (full numbers in the companion repository). Four additional checks leave every conclusion unchanged. (i) Re-estimating the CATE with an X-Learner (Künzel et al., 2019) raises absolute welfare but preserves the ranking (CATE Only 20,456, CATE $\times$ LTV 20,191). (ii) Drawing correlated covariates from a Gaussian copula ($\rho \in [0.3, 0.4]$) reproduces the independent-feature results (CATE Only 16,893, CATE $\times$ LTV 16,225). (iii) Re-scoring the headline table on the clip-adjusted effective treatment effect shows the nominal-welfare bias is modest (3.7–6.7% across strategies) and changes no ranking; the S-Learner's effective welfare (22,791) still dominates the best decomposed strategy's (16,044). (iv) Sweeping $\text{corr}(\tau, V)$ from -0.36 to $+0.20$ leaves value weighting's relative verdict at $\kappa = 0.30$ essentially unchanged (-1 to -2%), with the S-Learner leading throughout. Finally, an observational version of the regret diagnostic that replaces the oracle with a cross-fitted reference estimator reaches the same verdict (93% CATE share) and is therefore usable without oracle quantities, with the caveat that disagreement detects only *recoverable* error, not a value-model bias no available estimator can see.

8 Discussion

8.1 Three Scenarios, Three Regimes

The simulation yields a prescriptive guide for practitioners.

- **Well-specified CATE.** Calibration picks $g \approx 1$. Operators who want to exclude extreme-risk segments should choose their preferred g and read the welfare cost off the frontier (Table 9 and Figure 3).
- **Misspecified CATE.** Value-weighted targeting ($\hat{\tau} \cdot \hat{V}$) significantly beats CATE Only (6,955 vs. 6,700). The improvement is from V, not from g.
- **No CATE-risk interaction.** Value-weighted targeting dominates; calibration again picks $g \approx 1$.

The practical decision rule has two levels, and the first-order one is the estimator architecture, not the scoring weights. The estimated strategies fall into three families: the *decomposed* rule ($\hat{\tau} \cdot \hat{V}$ from separate T-Learner and value models, welfare 16,427), the *end-to-end* joint rule (a single model predicting the value-weighted outcome, 20,158; Section 7.7), and the *hybrid* rule (a joint treatment-effect model paired with a separate value model, 24,310; the "S-Learner" row of Table 3). The hybrid is the highest-welfare estimated strategy in this DGP, beating both the pure decomposition and the pure end-to-end model, and it is our default recommendation for an operator whose objective is welfare: use a joint treatment-effect model, but keep the value estimate separate so a well-specified $\hat{V}$ can still contribute. Reserve the decomposed rule for when transparency and component-wise control are required (below). The second-order choices concern the scoring weights within whichever architecture is chosen: use $\hat{\tau} \cdot \hat{V}$ rather than $\hat{\tau}$ alone at moderate-to-generous budgets but pure CATE at very tight budgets (Section 7.5); choose $g \neq 1$ only to encode non-welfare preferences such as risk control, regulatory compliance, or responsible gaming, and run the calibration procedure to price the chosen g against the identity.

Welfare vs. interpretability. The hybrid and end-to-end joint rules both beat the best decomposed strategy (CATE Only, 16,634 at this budget) by 46% and 21% respectively in this DGP, because a joint treatment-effect model can exploit cross-component interactions the decomposition cannot. The depth-matched ablation of Section 7.7 shows this gap is not an artefact of the tree-depth hyperparameter: at matched depth the gap is +53% to +74%, so the 46% figure is a lower bound on the joint-modelling advantage in this DGP rather than an upper one. The gap's magnitude is DGP-specific. The decomposed framework buys three things in exchange for this gap: independent auditability, component-wise updating, and transparent operator control via g. Of these, component-wise updating has a direct welfare consequence we can demonstrate: starting from the degraded pipeline of Section 7.3 (welfare 6,955) and improving only the treatment-effect model, welfare recovers to 16,427 (+136%) from a single-component update, whereas a joint model must be retrained end-to-end for the same gain. The broader auditability claim (inspecting and defending each component separately) is a governance benefit to which we assign no welfare number. We also flag a comparison that is easy to over-read. When login frequency is dropped from the *decomposed* CATE model, the decomposed strategies lose 60% of welfare while the S-Learner, which still sees all features, is unaffected (Section 7.3). This is not evidence that the joint model is more robust to missing information: dropping the same feature from the S-Learner degrades it by a comparable 64%. The joint model's advantage is that it uses a given feature set more effectively through cross-component interactions, not that it needs fewer features; both architectures are fragile to losing an informative one. In regulated settings such as online gaming, where operators must defend targeting decisions and different teams maintain different models, the auditability exchange can still be worthwhile, but

it should be made on governance grounds, with the substantial welfare cost (and the absence of a robustness advantage) stated plainly rather than assumed away.

8.2 Estimation in Practice

The composite score needs three estimates: $\hat{\tau}(x)$, $\hat{V}(x)$, and $\hat{R}(x)$. The framework is agnostic to the choice of estimator. Proposition 2 shows that errors in components with large magnitude, particularly V (which varies by orders of magnitude), are amplified by the other components and so contribute disproportionately to welfare loss.

Calibrating the parameters of g is essential. The default parameterisation costs 17% of welfare; calibration recovers the gap. For deployment we recommend the held-out validation procedure from Section 7.1, with the identity configuration included in the candidate grid.

The calibration procedure evaluates candidates against estimated welfare $\mathcal{W}(\hat{\pi}; \mathbf{1})$. Because Proposition 3 guarantees that $g \equiv 1$ maximises unrestricted welfare, any welfare-based calibration will pick $g \approx 1$ by construction. This is a consistency check on the theory, not a new finding. The primary practical role of calibration is therefore cost quantification. Each candidate g on the validation fold produces a point on the frontier; the operator reads off the welfare cost of their preferred preference configuration and decides whether the targeting control is worth it. The operator sets g from domain preferences; calibration tells them what g costs.

Plug-in versus doubly-robust calibration. The calibration above scores each candidate with the plug-in welfare $\hat{\tau} \cdot \hat{V}$ estimated on the estimation fold, which is biased toward rules aligned with the model's own estimation errors, so its clean selection of the identity (197 of 200 replications) could in principle be an artefact of scoring against those estimates. We therefore re-run calibration with a doubly-robust (AIPW) welfare estimate on the validation fold, using the known randomised propensity ($e = 0.5$) and T-Learner outcome models as the nuisance, exactly the AIPW machinery Section 8.3 recommends for deployment. Two things change and one does not. The near-identity selection rate collapses: AIPW selects the identity in only 14 of 200 replications, against 197 of 200 for the plug-in, because the doubly-robust score is higher-variance and its per-fold maximiser wanders across configurations. But the wandering does not buy welfare: scored on true (oracle) test welfare, the AIPW-calibrated rule underperforms the identity by 1,339 units on average (95% CI $[-1{,}704, -973]$, about 8%), whereas the plug-in-calibrated rule tracks the identity to within noise (-1 unit, CI $[-3, 1]$). So no g beats the identity on true welfare under either objective, which is what Proposition 3 requires; the plug-in's high identity-selection rate is partly a favourable artefact of its self-referential scoring, and a doubly-robust objective is the more honest but noisier selector. The headline conclusion, that calibration cannot discover a welfare-improving g, is robust to the scoring rule.

A natural follow-up question is whether a different calibration objective could allow the procedure to discover preference-useful g rather than always returning the identity. Candidates include a penalty-augmented welfare objective (estimated welfare minus an operator-supplied penalty on deviations from a preferred risk profile), a minimax-regret objective (selecting g that minimises the worst-case welfare loss across a plausible set of CATE misspecifications), or a constrained objective that caps risk exposure at an operator-chosen ceiling. Each of these changes the calibration problem from "find the welfare-optimal g" (which Proposition 3 already answers) into "find the best trade-off against a separately specified criterion", allowing non-identity solutions to arise. We do not pursue these here because they require additional operator input beyond the scope of the present framework, but they are natural extensions of the calibration machinery.

Remark 13 (Churn proxy vs. structural churn probability)**.** *The churn model in the simulation is trained to predict $1 - Y_i \mid W_i = 0$, the control-group non-retention indicator. This estimates*

$P(Y(0) = 0 \mid X)$, which is correlated with but not identical to the structural churn probability p_i^{churn} generated from the logistic model in the DGP. The distinction matters because the risk-modulation function g operates on estimated churn: if the churn model captures different information than structural churn risk, the effective modulation function differs from the intended one. In practice, churn models are typically richer than the binary proxy used here, incorporating longitudinal behavioural signals; the simulation understates both the difficulty and the potential value of churn estimation.

8.3 Evaluation in Practice

The simulation evaluates strategies using oracle (ground-truth) potential outcomes. In deployment those are never available: we only ever observe one of the two outcomes per individual (treated or not), which means realised welfare has to be *estimated*, not measured directly. The standard approach is *off-policy evaluation*: estimating what welfare a proposed policy would have produced using data collected under a different (typically randomised) policy. Two families of estimators are suitable. The first is the *augmented inverse propensity weighting* (AIPW) estimator (Dudik et al., 2011), which combines a model of the outcome with the inverse of the treatment-assignment probability (the *propensity score*). AIPW is *doubly robust*: it produces a valid welfare estimate if either the outcome model or the propensity model is correctly specified (both can be wrong, but only one at a time). The second is *counterfactual risk minimisation* (Swaminathan and Joachims, 2015), which reframes policy learning as minimising an importance-weighted version of the empirical loss on historical data ("importance-weighted" meaning each observation is reweighted by how likely the proposed policy would be to assign its observed treatment, relative to the policy that actually collected the data). The treatment-effect estimates already used inside the composite score can double as the outcome model in AIPW, so the targeting and evaluation pipelines share component estimates. In deployment we recommend AIPW-based welfare monitoring combined with a randomised holdout group that receives random treatment assignments, to maintain valid propensity scores over time. This recommendation interacts with the responsible-gaming safeguards of Section 6.2 in a way that a gambling operator cannot ignore. The exclusion set $\mathcal{E}$ and the harm gate remove individuals from the eligible pool *before* ranking; those individuals must also be removed from the randomised holdout, because randomising an at-risk or self-excluded player into treatment is precisely what the safeguards forbid, and in several jurisdictions is prohibited outright. The holdout therefore provides valid propensities only for the *eligible* population, and welfare estimates from it generalise only to that population, not to the excluded tail. Operators should treat the propensity model, the exclusion set, and the harm gate as a single coupled system: a player's eligibility can change between campaigns (Section 6.2), so holdout membership must be re-derived each round rather than held fixed.

8.4 Assumptions and Scope

The multiplicative welfare function (Definition 3) requires that the operator's valuation factor into the product of treatment effect, outcome value, and risk modulation (it is *separable* in the three components). We adopt separability deliberately, for two reasons, and we are explicit that our optimality results are conditional on it rather than derived from more primitive axioms. First, separability is the form the heuristic the paper is about already takes: practitioners rank by $\hat{\tau} \cdot \hat{V}$, a product, so the multiplicative welfare function is the objective implicitly revealed by existing practice, not an arbitrary imposition. Second, separability is exactly what buys the decomposition, auditability, and component-wise update properties that distinguish this framework from a single joint model; a non-separable objective would not merely complicate the algebra, it would dissolve the framework's reason to exist. The honest reading is therefore that separability is a scope condition: where it holds, the composite-score rule is welfare-optimal and

the regret decomposition applies; where it fails, the framework should not be used.

This rules out non-separable preferences. As a concrete example: suppose an operator's willingness to take risk exposure varies with customer value; they are more willing to target stable (low-churn) high-value customers than stable low-value customers, because the returns when retention goes through are larger in the former case. The effective preference weight then depends jointly on R and V, so the preference factor becomes $g(R, V)$ rather than $g(R)$, and the composite score no longer factors into a product of three univariate components. Such preferences require a more general welfare function than the one used here, and the decomposition, auditability, and component-wise update properties that motivated our framework would weaken accordingly; an operator who holds them should treat that as a signal to abandon the decomposed framework rather than to patch it.

The binary outcome assumption ($Y_i \in \{0, 1\}$) restricts the framework to the *extensive margin*: the outcome is whether a thing happened (the customer retained, the patient recovered), not how much. Extension to continuous outcomes ("how much revenue did the customer generate in the post-treatment window") is straightforward and does not change any of the results. With $Y \in \mathbb{R}$, the CATE $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$ is simply the expected continuous-valued gain, and V may be redundant (or may be interpreted as a multiplier that re-weights the continuous outcome to a common welfare unit, e.g., monetising QALYs). The composite score $\tau(x) \cdot V(x) \cdot g(R(x))$ and all three propositions go through with obvious modifications. In many operational gaming settings the continuous outcome (expected value over a retention window) is the quantity operators actually optimise; the binary framing here is chosen for clarity of exposition and because retention is the standard reference outcome in the uplift literature.

The pre-treatment requirement for the auxiliary components (Remark 2) is a real restriction. In sequential deployment, both churn probability and outcome value at a later period will reflect the effects of prior treatments, which means they are no longer strictly pre-treatment quantities; this introduces a statistical problem called *endogeneity* (the auxiliary variables become correlated with the unobserved drivers of the new treatment's effect), and the causal interpretation of the composite score breaks down.

We assume treatment is equally costly for every individual; if costs differ (e.g., bonuses of different sizes), the appropriate ranking is by the net-benefit score $s_i - c_i$ rather than s_i, where c_i is the per-individual treatment cost. This homogeneous-cost assumption is a genuine limitation for bonus targeting specifically, where the realised cost is *response-dependent*: a retention bonus costs the operator only when it is redeemed, and its size then depends on subsequent wagering, so c_i is itself a function of the player's response rather than a fixed pre-treatment quantity (Haupt and Lessmann, 2022). When c_i depends on the treatment response, the net-benefit correction $s_i - c_i$ no longer factors cleanly out of the welfare-optimal rule, because cost and effect are jointly determined; Haupt and Lessmann (2022) treat exactly this case and derive a profit rule under response-dependent costs. Extending the composite score to response-dependent costs (rather than the fixed c_i handled here) is the most consequential domain-specific extension for bonus targeting, and we leave a simulation of it to future work.

The SUTVA assumption (Assumption 1(iii)), no interference between individuals, may be violated in settings where individuals interact: gaming platforms with guilds, multiplayer modes, or shared play sessions; clinical trials where patients influence each other; online services with social features. If treating individual i affects the outcome of individual j, the individual-level treatment effect τ_i is no longer well-defined: it depends on whether i's peers are also treated. This kind of spillover is known in the causal-inference literature as *network interference* (each individual's outcome depends on the treatment assignments of their network neighbours, not just their own treatment). Extending the framework to account for network interference is a direction for future work.

Remark 14 (Terminology: risk modulation vs. risk aversion)**.** *The "risk-modulation" function g encodes covariate-dependent targeting weights based on an observable auxiliary variable (churn probability), not risk aversion in the decision-theoretic sense of Manski and Tetenov (2007). The function g is better understood as a domain-informed reweighting of the welfare integrand.*

The multiplicative CATE decomposition $\tau(x) = \tau_0(x) \cdot m(R(x))$ assumed in Remark 12 is an approximation. Additive interactions and threshold effects are not well-served by the multiplicative framework; the operator should assess whether the domain structure supports multiplicative moderation before deploying g.

8.5 Extensions

Natural directions for future work include: (i) multi-armed and sequential treatment with bonus sizing (selecting from a portfolio of intervention types); (ii) finite-sample regret bounds for the three-component product score analogous to the $O(n^{-1/2})$ rates of Athey and Wager (2021); and (iii) fairness constraints within the budget-constrained optimisation.

9 Conclusion

Proposition 1 places value-weighted uplift targeting inside the budget-constrained empirical-welfare-maximisation framework: under the multiplicative-separable welfare function of Definition 3 and a budget constraint, the welfare-maximising deterministic rule ranks by $\tau \cdot V$ (or, with operator preferences, by $\tau \cdot V \cdot g(R)$). The decision-theoretic merit of value weighting is established in prior marketing work; the multiplicative separability that lets it nest into policy learning is itself a modelling assumption, not a derivation, and its scope is discussed in Section 8. The paper's two more durable contributions are the regret decomposition and the operating-regime map it explains. The regret decomposition (Proposition 2, Remark 9) localises welfare loss to individuals misranked near the threshold and attributes it to a specific component's estimation error, so it tells an operator which of the three estimates to improve first. The operating-regime map is the simulation's main empirical finding: value weighting matters most exactly where it is most needed. Under CATE misspecification, $\hat{\tau} \cdot \hat{V}$ significantly beats CATE Only (6,955 vs. 6,700) because the value model retains features the uplift model drops; this holds only when the dropped features also drive value, and an adversarial misspecification orthogonal to value removes the benefit (Section 7.3). The effect's direction is structural; its magnitude is DGP-specific. Value weighting's advantage is also budget-dependent: at moderate-to-generous budgets it matches or exceeds CATE Only, but at very tight budgets pure CATE is stronger because value weighting dilutes the uplift signal.

Adding the operator-configurable preference weight g extends the framework to encode domain preferences (excluding extreme-risk segments, responsible-gaming considerations, regulatory constraints). Proposition 3 prices this: any $g \neq 1$ weakly reduces welfare relative to $g \equiv 1$. The simulation's calibration procedure reproduces the identity in every scenario, which is what Proposition 3 implies it must do; the substantive role of calibration is to produce the welfare-cost schedule across different preference configurations. The operator's role is not to discover a welfare-improving g (there is none, by construction); it is to decide which preferences are worth what welfare cost.

Joint modelling of the treatment effect beats the decomposition on welfare in this DGP, and the strongest estimated strategy is neither the pure decomposition nor the pure end-to-end joint model but the *hybrid* between them: a joint treatment-effect model paired with a separate value model (24,310, against 20,158 for the end-to-end model and 16,427 for the decomposition). The depth-matched ablation, the end-to-end joint model, and the effective-welfare re-scoring (all in Section 7.7) confirm the joint-modelling advantage is not an artefact of tree depth, the borrowed

value estimate, or the nominal-welfare convention. When a single flexible model is available and interpretability is not required, the hybrid architecture wins on welfare, and a well-specified separate value model still earns its place within it. The decomposed framework earns its keep where transparency, auditability, and component-wise control matter, such as regulated online gaming, where the piecewise linear g gives operators five interpretable knobs with operational meanings.

We close by stating the paper's posture plainly, because it is as much a caution as a method. An operator who optimises welfare alone should, on this evidence, reach for the hybrid architecture (a joint treatment-effect model with a separate value model) and not for the decomposed framework; the operator-preference weight g should be set from domain preferences and priced, never tuned for welfare, because it provably cannot improve it; and value weighting should be switched on at moderate-to-generous budgets and off at tight ones. What the framework offers is not a higher-welfare estimator but a common objective in which these statements can be made precisely, the regret decomposition described above, and a transparent, auditable structure whose welfare cost is now quantified rather than assumed. Whether that transparency is worth its welfare cost is a governance judgement we make explicit rather than resolve.

The main limitation is the absence of real-data validation: the qualitative findings are stable across the structural variations we test (two moderation shapes, a correlation sweep between effect and value, and a heavy-tailed value distribution), but deployment performance ultimately depends on the quality of the component estimates in the field. The public uplift benchmarks (e.g., Criteo, Hillstrom) are only a partial remedy here: they supply real covariates and randomised treatment but not the two quantities the framework turns on, a per-individual outcome value V and a risk covariate R, so a semi-synthetic study on them would still simulate the very components (V, R) whose interaction with τ is the object of study. Such a study would validate the CATE estimation but not the value-weighting or preference-pricing claims; genuine validation requires operator data with observed lifetime value and a churn/risk signal. For production deployment, the framework should be paired with off-policy welfare evaluation (e.g., AIPW estimators) on observed data. Validating the framework on real targeting data, extending it to multi-armed and sequential treatment, and deriving finite-sample regret bounds for the three-component product score are natural directions the analysis here opens up rather than commitments we make.

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Conflict of Interest and Disclosure

The author is employed by Allwyn, an operator in the gaming and lottery sector, and the framework studied here is directly relevant to that sector. This is a potential conflict of interest and is disclosed in full. The views expressed are the author's own and do not necessarily represent the position of Allwyn. The research received no specific external grant funding. The framework is presented with explicit responsible-gaming safeguards (Section 6.2), and the author draws an explicit distinction between operator welfare (the econometric quantity the method optimises) and player welfare, cautioning that welfare-optimal targeting in the econometric sense is not the same as ethically optimal targeting in a harm-sensitive domain.

Ethics Statement

This study involves no human participants, no personal data, and no experiments on individuals: all results are produced by a synthetic, fully specified simulation (Section 7), and no real player, customer, or patient data was used. No ethics-board approval was therefore required. Because the application domain (promotional targeting in online gambling) carries real potential for harm, the paper foregrounds that risk rather than abstracting it away: it documents that value-weighted targeting can concentrate incentives on the most-harmed players (Section 6.2), treats a vulnerability-based exclusion set and a harm gate as preconditions for any responsible deployment, recommends conservative defaults (a zero floor on the highest-risk segment), and notes that direct promotion to identified at-risk players may itself be restricted in some jurisdictions. The framework should not be deployed in a gambling setting without the responsible-gaming safeguards described in Section 6.2 and the validated harm-measurement and regulatory infrastructure they depend on.

Data and Code Availability

This study uses no empirical data; all reported quantities are generated by simulation. The complete simulation code, the revision-additions code (asymmetric-moderation and correlated and heavy-tailed value DGPs, the harm-gate and hard-constraint analyses, the component-update and end-to-end-joint comparisons, the clip-adjusted welfare re-evaluation, and the observational diagnostic), the fourth-revision additions (budget-sensitivity inference and the crossover budget $\kappa^\star$ with bootstrap CIs, the doubly-robust AIPW calibration check, and the adversarial-misspecification DGP), the fifth-revision addition (the like-for-like joint-model comparison that drops the same feature from the S-Learner), the data-generating process specification, the fixed random seeds ($42+7r$ for replication r), and the analysis scripts that regenerate every table and figure are available in the companion repository. The output files themselves are not versioned; because every replication uses a fixed seed, the entire study is deterministic and regenerable by running the released scripts under the pinned dependencies. For peer review the repository is provided as anonymised supplementary material; a public URL and archival DOI will be included in the camera-ready version.

A Proofs of Propositions 1 and 2

Proof of Proposition 1 (optimal risk-modulated treatment rule)

Proof. The problem is to choose the rule π that maximises

$$\max_{\pi \in \Pi_\kappa} \ \mathbb{E}\big[\pi(X_i) \cdot \tau(X_i) \cdot V_i \cdot g(R_i)\big]$$

subject to the budget cap. The standard way to handle a budget constraint in an optimisation problem is the method of Lagrange multipliers: we attach a non-negative number $\lambda \geq 0$, called a *Lagrange multiplier*, to the budget constraint and add it to the objective. Doing so gives the *Lagrangian*

$$\mathcal{L}(\pi, \lambda) = \mathbb{E}\big[\pi(X_i)(\tau(X_i) \cdot V_i \cdot g(R_i) - \lambda)\big] + \lambda\kappa.$$

The multiplier λ is a bookkeeping device at this stage; at its optimum value it coincides with the score threshold c^*, and Remark 5 gives the economic interpretation of that threshold as the shadow price of the budget. Reading the Lagrangian, for any fixed λ the decision-maker is free to set $\pi(x) \in \{0, 1\}$ individual by individual. Each individual's contribution to the Lagrangian is $\pi(x)$ times the quantity $\tau(x) \cdot V(x) \cdot g(R(x)) - \lambda$. Because $\pi(x)$ is binary, the individual-level

contribution is maximised by setting $\pi(x) = 1$ whenever that quantity is positive and $\pi(x) = 0$ otherwise. This gives the threshold rule

$$\pi_\lambda(x) = \mathbf{1}\{\tau(x) \cdot V(x) \cdot g(R(x)) > \lambda\}.$$

(This kind of argument, where the maximiser is pinned down individual by individual rather than globally, is called a *pointwise* argument.) At the boundary $\tau(x)V(x)g(R(x)) = \lambda$ either choice of $\pi(x) \in \{0, 1\}$ is equally optimal for the Lagrangian; Assumption 2(i) makes this boundary set a measure-zero event, so the strict vs non-strict inequality in the threshold rule has no welfare consequence. Setting $c^* = \lambda^*$, where λ^* is chosen to exactly exhaust the budget ($\mathbb{E}[\pi_{\lambda^*}(X_i)] = \kappa$), yields the optimal rule. This characterisation presumes the budget binds, which requires that at least a fraction κ of the population has a strictly positive composite score: $P(s(X) > 0) \geq \kappa$. When this holds, the existence and uniqueness of $\lambda^* > 0$ (equivalently $c^* > 0$) follow from the continuity condition in Assumption 2. When it fails, that is, when fewer than a fraction κ of individuals have a positive score, the constraint slacks: by complementary slackness $\lambda^* = 0$, the optimal rule treats every individual with $s(X) > 0$ (and no one with $s(X) \leq 0$, since treating a non-positive-score individual cannot raise welfare), and $c^* = 0$ is no longer unique on the atom at zero. This degenerate case is exactly what an aggressive preference weight with a wide $g \equiv 0$ exclusion zone can induce. The remaining step is to verify that solving the Lagrangian gives the solution to the original constrained problem, a property called *strong duality*. Strong duality holds for the *convex relaxation* of the problem, in which $\pi(x)$ is allowed to take any value in $[0, 1]$ rather than just $\{0, 1\}$: the relaxed problem has a linear objective and a linear constraint, and any rule that treats strictly fewer than a fraction κ of the population is strictly feasible, so Slater's condition applies. The relaxed optimum is attained by a binary-valued rule almost surely: for each x the pointwise-optimal $\pi(x) \in [0, 1]$ is 1 when the integrand is positive and 0 when it is negative, and the measure-zero boundary noted above makes any intermediate value $\pi(x) \in (0, 1)$ irrelevant. The relaxed solution therefore coincides with the solution to the original binary problem. The overall argument (a threshold rule on an objective integrand maximises the objective under a single budget constraint) is the standard Neyman-Pearson logic used in statistical testing and selection problems, applied by Kitagawa and Tetenov (2018, Theorem 1) to the CATE-only objective and extended here to the three-factor product $\tau(x) \cdot V(x) \cdot g(R(x))$. □

Proof of Proposition 2 (welfare regret decomposition)

Proof. By linearity of expectation,

$$\begin{aligned}\mathcal{R} &= \mathbb{E}\big[\pi^*(X_i) \cdot s(X_i)\big] - \mathbb{E}\big[\hat{\pi}(X_i) \cdot s(X_i)\big] \\ &= \mathbb{E}\big[(\pi^*(X_i) - \hat{\pi}(X_i)) \cdot s(X_i)\big].\end{aligned}$$

For any individual i not in $\mathcal{M}$, the two rules agree, so $\pi^*(X_i) - \hat{\pi}(X_i) = 0$ and that individual contributes nothing to the expectation. The regret therefore reduces to an expectation over the misclassified set alone. Taking absolute values and using $|\pi^* - \hat{\pi}| \leq 1$ (the difference of two binary indicators is between -1 and 1) gives (7). For (8), decompose $\mathcal{M}$ into false negatives $\mathcal{M}^+ = \{\pi^* = 1, \hat{\pi} = 0\}$ and false positives $\mathcal{M}^- = \{\pi^* = 0, \hat{\pi} = 1\}$, and write

$$\mathcal{R} = \mathbb{E}[s(X_i) \cdot \mathbf{1}\{\mathcal{M}^+\}] - \mathbb{E}[s(X_i) \cdot \mathbf{1}\{\mathcal{M}^-\}].$$

When both rules exhaust the budget, $\mathbb{E}[\pi^*] = \mathbb{E}[\hat{\pi}] = \kappa$, so $\mathbb{E}[\pi^* - \hat{\pi}] = 0$; since $\pi^* - \hat{\pi} = \mathbf{1}\{\mathcal{M}^+\} - \mathbf{1}\{\mathcal{M}^-\}$, this gives the equal-mass identity $\mathbb{E}[\mathbf{1}\{\mathcal{M}^+\}] = \mathbb{E}[\mathbf{1}\{\mathcal{M}^-\}]$ (equality of expected mass, not of cardinality). Consequently $c^* \cdot (\mathbb{E}[\mathbf{1}\{\mathcal{M}^+\}] - \mathbb{E}[\mathbf{1}\{\mathcal{M}^-\}]) = 0$, so subtracting and adding $c^* \cdot \mathbf{1}\{\mathcal{M}^\pm\}$ leaves the expectation unchanged and yields

$$\mathcal{R} = \mathbb{E}[(s(X_i) - c^*) \cdot \mathbf{1}\{\mathcal{M}^+\}] + \mathbb{E}[(c^* - s(X_i)) \cdot \mathbf{1}\{\mathcal{M}^-\}].$$

On $\mathcal{M}^+$ the oracle treats, so $s(X_i) \geq c^*$; on $\mathcal{M}^-$ it does not, so $s(X_i) \leq c^*$. Both terms are therefore non-negative and combine to give $\mathcal{R} = \mathbb{E}[|s(X_i) - c^*| \cdot \mathbf{1}\{i \in \mathcal{M}\}]$. □

B The Piecewise Linear Risk-Modulation Function

This appendix gives the formal definition of the five-parameter risk-modulation function g used in Section 6 and its regularity properties. The functional form is an implementation choice; the operational reading of the parameters and the default values θ_0 are in the body.

Definition 5 (Piecewise linear risk-modulation function)**.** *The risk-modulation function* $g : [0,1] \to [0,1]$ *is defined as*

$$g(p;\,\theta) = \begin{cases} 0 & \text{if } p < p_L, \\ \dfrac{p - p_L}{p_{ML} - p_L} & \text{if } p_L \le p < p_{ML}, \\ 1 & \text{if } p_{ML} \le p \le p_{MH}, \\ 1 - (1-\phi)\,\dfrac{p - p_{MH}}{p_H - p_{MH}} & \text{if } p_{MH} < p < p_H, \\ \phi & \text{if } p \ge p_H, \end{cases} \tag{19}$$

parameterised by $\theta = (p_L, p_{ML}, p_{MH}, p_H, \phi)$ *with* $0 \le p_L < p_{ML} \le p_{MH} < p_H \le 1$ *and* $\phi \in [0,1]$.

Remark 15 (Properties of the piecewise linear g)**.** *The function* g *defined in* (19) *satisfies: (i)* g *is continuous on* $[0,1]$*; (ii)* g *is non-decreasing on* $[0, p_{MH}]$ *and non-increasing on* $[p_{MH}, 1]$*; (iii)* $g(p) = 0$ *for* $p < p_L$ *and* $g(p) \ge \phi$ *for all* $p \ge p_H$*; on the ascending ramp* $[p_L, p_{ML})$*,* $g(p)$ *can be below* ϕ *when* $\phi > 0$*; (iv)* g *is Lipschitz continuous, so expectations involving it are well-behaved. The five-parameter form is an implementation choice, not part of the contribution; any bounded, measurable weight would serve, and the parameters are chosen for their direct operational reading (Section 6) rather than for approximation-theoretic reasons.*

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